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CHAPTER ONE

FOREIGN EXCHANGE MARKET

Learning Objectives

- Meaning of foreign exchange market
- Learn who participates in foreign exchange markets and why
- Discussing exchange rate regime
- Defining purchasing power parity

1.1. Foreign Exchange Market

What is foreign exchange market?

Foreign exchange is the **trading of currencies**. The foreign exchange market is not a single place like the NY Stock Exchange (NYSE). Instead, the foreign exchange market refers to the activities of major international banks that engage in currency trading. These banks act as intermediaries between the true buyers and sellers of currencies (i.e., government, business, and individuals). These banks will hold foreign currency deposits and stand ready to exchange these for domestic currency upon demand. It is a widely decentralized 24-hour-a-day market, made up of banks and traders communicating electronically. The retail market is b/n individuals, nonfinancial companies, nonbank financial institutions includes investment banks, mortgage lenders, money market funds, insurance companies, etc, and other customers of banks. The wholesale or interbank market is the trading between banks. This accounts for 60% or more of the total trading. The foreign exchange market is thus the market in w/c individuals, firms and banks buy and sell foreign currency.

1.2. Characteristics of foreign exchange market and its participant

The following points highlight the top seven characteristics of foreign exchange market. the characteristics are: 1. most liquid market in the world 2. most dynamic market in the world 3. twenty-four hour market 4. market transparency, 5. international network of dealers 6. most widely traded currency is the dollar 7. “over-the-counter” market with an “exchange-traded” segment.

1. Most liquid market in the world

Currency spot trading is the most popular FX instrument around the world, comprising more than 1/3 of the total activity. It is estimated that spot FX trading generates about \$1.5 trillion

a day in volume, making it the largest most liquid market in the world. Liquidity has its advantages, the primary one being no manipulation of the market. Thin stock and futures markets can easily be pushed up or down by specialists, market makers, commercials, and locals. Spot FX on the other hand takes real buying/selling by banks and institutions to move the market. Any attempted manipulation of the spot FX market usually becomes an exercise in futility.

2. Most dynamic market

Foreign exchange market is the most dynamic market in the world regardless of which instrument you are trading like stocks, government bonds, Ethiopian government treasuries, foreign exchange , or any of the countless others the attributes that determine the viability of a market as an investment opportunity remain the same.

3. It is a twenty-four hour market

The twenty-four hour market means that exchanges rates and market conditions can change at any time in response to developments that can take place at any time. It also means that trades and other market participants must be alert to the possibility that a sharp move in an exchange rate can occur during an off hour, elsewhere in the world.

4. Market transparency

Price transparency is very high in the FX market and the evolution of online foreign exchange trading continues to improve this, to the benefit of traders. One of the biggest advantages of trading foreign exchange online is the ability to trade directly with the market maker. A reputable forex broker will provide traders with streaming, executable prices. It is important to make a distinction between indicative prices and executable prices.

5. International Network of Dealers

The market is made up of an international network of dealers. The market consists of a limited number of major dealer institutions that are particularly active in foreign exchange, trading with customers and (more often) with each other. Most, but not all, are commercial banks and investment banks. These dealer institutions are geographically dispersed, located in numerous financial centers around the world. Wherever located, these institutions are linked to, and in close communication with, each other through telephones, computers, and other electronic means.

6. Most widely traded currency is the Dollar

The dollar is by far the most widely traded currency. According to the 1998 survey, the dollar was one of the two currencies involved in an estimated 87 percent of global foreign exchange transactions, equal to about \$1.3 trillion a day. In part, the widespread use of the dollar reflects its substantial international role as – “investment” currency in many capital markets, “reserve” currency held by many central banks, “transaction” currency in many international commodity markets, “invoice” currency in many contracts, and “intervention” currency employed by monetary authorities in market operations to influence their own exchange rates.

Examples of currencies used by different

Country	Currency	Symbol	ISO Code
US	Dollar	\$	USD
UK	Pound	£	GBP
Canada	Dollar	C\$	CAD
Germany	Deutsche Mark	DM	DEM
France	Franc	FF	FRF
Mexico	Peso	Ps	MXP
Japan	Yen	¥	JPY
Euro zone	Euro	€	
Ethiopia	Birr	Br	

1.3. The functions of foreign exchange market

The following are the main functions of foreign exchange market, which are actually the outcomes of its working;

- To facilitate currency conversion
- Provide instruments to manage foreign exchange risk such as foreign exchange
- Allow investors to speculate in the market for profit

- It transfer funds or purchasing power from one nation & currency to another

1.3.1. Scope of the foreign exchange market

About half of the daily foreign exchange trading is done b/n banks in London and New York. In 2001, the trading volume was about \$1.2 trillion per day. (The NYSE turned over about \$42 billion per day.) Most of the trading involves U.S. currency. Sometimes the intent is to trade one foreign currency for another, & the U.S. currency is only involved as an intermediate step. When this is done, the dollar is called a vehicle currency.

1.3.2. Exchange rate

The **exchange rate** is the price of one country's money in terms of another country's money. The exchange rate can be given as the **price of the foreign currency in terms of the domestic currency**—this is the usual way, & the way we'll use in this course— (domestic currency/foreign currency) or as the price of the domestic currency in terms of the price of the foreign currency (Foreign currency/domestic currency).

The “**spot**” **exchange rate** is the price for immediate exchange. (Immediate usually means within two working days.) This amounts to about 33% of all trading. The “forward” exchange rate is the price for exchange to take place at some specific time in the future, often 30, 90, 180 days. This amounts to about 11% of all trading.

A “**swap**” is a “**package trade**” that includes both a spot exchange of two currencies & a contract to the reverse forward exchange a short time later. This is useful when the parties to the swap have only a short-term need for the currency. This amounts to about 56% of all trading.

Nominal effective exchange rate (NEER) –is an unadjusted weighted average rate at w/c one country's currency exchanges for a basket of multiple foreign currencies. It is the amount of domestic currency needed to purchase foreign currency. It is exchange rate without adjusting for inflation. **Example:** the **nominal effective exchange rate** of the euro for France is a weighted mean (with the weighting being specific to France) of the exchange rates of the euro against the currencies of the computing countries in a given zone, i.e euro zone

The **real effective exchange rate (REER)** is the weighted average of a country's currency in r/n to an index or basket of other major currencies. The weights are determined by comparing the

relative trade balance of a country's currency against that of each country in the index. REER is determined by taking the average of the bilateral exchange rates b/n one nation & its trading partners & then weighting it to take into account the trade allocation of each partner. **Example** let's say the U.S. had a foreign trading relationship with only three parties: the euro zone, Great Britain, and Australia. That means the U.S. dollar has a trading relationship with the euro, the British pound, and the Australian dollar. The **real effective exchange rate** of the euro for France includes the exchange rate but also the ratio of France's export prices to those of competing countries in the zone under consideration. Which indicates it is adjusted for inflation.

1.3.3. Determinants of foreign exchange rate

There are so many factors that influence exchange rates. But five factors are considered most important in influencing foreign exchange rates. These are;

- **Changing inflation:** A country with a lower inflation rate has greater purchasing power against other currencies and so displays rising currency value. Higher rate of inflation obviously lowers currency value. However, a currency may sometimes strengthen when inflation rises because of expectations that the central bank will raise short-term interest rates to combat rising inflation.
- **Changing interest rate:** Inflation, interest rates and exchange rates are closely related to each other. Central banks forge interest rates to influence exchange rates and inflation which directly affect the inflation and forex rates. Higher interest rates benefit investors to attract more foreign investors. This results in an increase in a foreign capital with the country and increased foreign exchange rates.
- **Relative income levels:** The third factor influencing the exchange rate is relative income level because income level affects the amount of import demanded and the exchange rate. Let, between the US and UK, UK income level increases and US income level remains unchanged. As a result demand curve will shift upward because of the increase in UK income and increased demand for US goods. But the supply schedule is not expected to change. For this reason, there is an increase in the equilibrium exchange rate.
- **Government Controls and change in expectations:** Government can control equilibrium exchange rates by many ways. There are so many factors, such as foreign exchange barriers, foreign trade barriers, Intervening transactions etc., in the foreign

exchange market that government can impose to affect the exchange rate (Madura, 2010). The last factor that influences the exchange rate is expectations of future exchange rates. Like all financial markets, foreign exchange markets react to the news that can create future effects. The effective can either increase or decrease the exchange rate (Eitrman, 2013). Let news spreads in the US that inflation will increase in the recent future. It will make traders sell US dollars and supply will increase. But, demand will have no impact. As a result, supply will be decreased and this will result in a decreased exchange rate. But if news spread that there will be deflation in the US, traders will buy US dollars as they will try to purchase all the US dollars from the market, increasing US dollar demand. But there will be no change in supply. This will increase equilibrium exchange rate.

Why exchange rate?

It has strong influence on current account & other macroeconomic variables. As it is the price one currency in terms the other, it governs the asset price. Facilitate the flow of asset from one place to another. It allows comparing the price of goods & services produced at d/t places/countries.

Types of Exchange rate markets

Foreign exchange market is broadly divided into: **spot exchange market & forward exchange market**.

- **Spot Market Transaction**

The **spot market transaction** determines the spot exchange rate(S). This indicates the prices of currencies for immediate transaction. It may take 1-2 days to settle the payment. The term spot is used here to imply short time period only. **Example:** British firm buys a U.S. product from a U.S. firm, which requires payment in U.S. dollars (\$). The British firm contacts its bank, gets a quote on the dollar-pound exchange rate, and approves it. The British firm instructs its bank to take pounds from its checking account, convert these to dollars, & transfer the amount to the U.S. producer. The British bank instructs its “correspondent” bank in New York to take U.S. dollars from its account and pay the U.S. producer by transferring them to the producer’s bank.

- **Forward Exchange Market**

The **forward exchange market** refers to the foreign exchange deal /contractual agreement / for sale and purchase of foreign currency at some future date. **Forward rate, (F)**, is that rate negotiated today at which foreign exchange can be bought and sold for delivery sometime in the future. The most popularly traded forward contract has a maturity of 90 days and contracts beyond 1 year are relatively scarce. Forward contracts are generally negotiated between an individual – for example, a private customer or commercial organization –and a bank & the individual has to take delivery of the contract on the specified date. It is one way of hedging against exchange rate risk. **Example:** suppose that a US company buys radio from the Japan Sony company expecting a shipment of the Radio in a month, for which it needs to pay yen. Could the company wait until the shipment to buy the yen to pay Sony? But it does not know what will happen to value of that yen in meantime. If yen appreciates a lot, the \$ price US Company has to pay to get the radio could change a lot. i.e.... the US company need to pay more dollars per yen. To avoid this risk, the company can arrange for currency trade ahead of time to be executed later at a set exchange rate.

- **Automated systems in foreign exchange market**

SWIFT (the Society for Worldwide Interbank Financial Telecommunications) it is Electronic System with over 2,000 member banks in almost 200 countries. About 4 million transactions per day.

CHIPS (the Clearing House International Payments System) clears dollar transfers among member banks, includes all major, internationally-active banks the ease of use of CHIPS makes the dollar convenient as a vehicle currency clears over \$1 trillion dollars per day.

Interbank Trading: Conducted by brokers and traders. Traders work in trading rooms with computer terminals & telephones. Traders get to know their counterparts at other banks very well

Interbank “rates” are quoted for amounts of \$1 million or more, & a trader may handle millions of dollars of foreign exchange in a matter of minutes. The volumes are so large, that they often refer to \$1 million of exchange as “one dollar”.

1.4. The demand for and the supply of foreign exchange

The exchange rate of the foreign exchange (dollar) will be determined by the intersection of the supply & demand for dollar on the foreign exchange market in the world financial markets. The **Demand for foreign exchange**; the demand for currency in the foreign exchange market is a derived demand. It is derived from the demand for US product. i.e..., dollar is not demanded because it has intrinsic value by itself, but rather because of what it can buy. **Example**; to derive hypothetical demand for dollar let us assume that US is exporting country and Ethiopia being importer.

Table 1.2. Demand for dollar

Price of US Export good in USD(1)	Exchange rate(2)	Price of US export in birr (3)(1x2)	Quantity of US export (4)	Demand for USD(5)=(1)(4)
10	9.20 birr	92	1500	15000
10	9.30	93	1300	13000
10	9.40	94	1100	11000
10	9.50	95	900	9000
10	9.60	96	700	7000
10	9.80	98	500	5000
10	10.00	100	300	3000

If **dollar appreciate** against birr, the price of the US export increase and this leads to a lower quantity of US exports thus reduce demand for dollar.

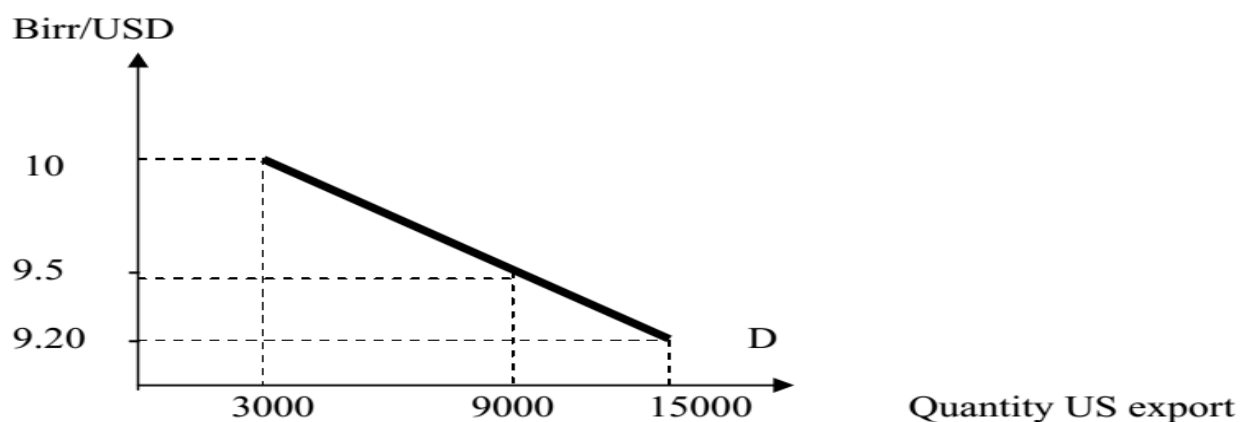


Fig. 1.1. Demand for dollar

Note that any factor that affects the demand for US export affects the demand for dollar. Such as: change in ETH income, change in the price of ETH goods (which can substitute imports from US) change in ETH tastes in favor of US goods .All the above factors cause shift in demand for dollar.

Supply of foreign exchange: If dollar appreciates, the US export become expensive to the Ethiopian importers, on the other hand, Ethiopian export becomes cheap to the US residents. This leads to increase in demand for Birr, leading to rise in supply of USD. This show an upward sloping supply schedule for USD. Consider the following table.

Table 1.3. The Supply of dollar

Price of ETH export goods in birr (1)	Exchange rate birr/USD (2)	Price of ETH export in USD (3) = $(\frac{1}{2})$	Quantity of ETH export (4)	Demand for birr (1)x(4) (5)	Supply of USD (6) = $(\frac{5}{2})$
100	9.20	10.9	300	30000	3260.90
100	9.30	10.75	500	50000	5376.34
100	9.40	10.64	700	70000	7446.80
100	9.50	10.53	947.7	94770	9975.79
100	9.60	10.42	1200	120000	12500.00
100	9.80	10.21	1350	135000	13775.50
100	10.00	10	1500	150000	15000.00

Supply curve of foreign exchange slope upwards due to positive relationship between supply for foreign exchange and foreign exchange rate. In Fig. 1.2, supply of foreign exchange (US Dollar) and rate of foreign exchange have been shown on the X-axis and Y-axis respectively.

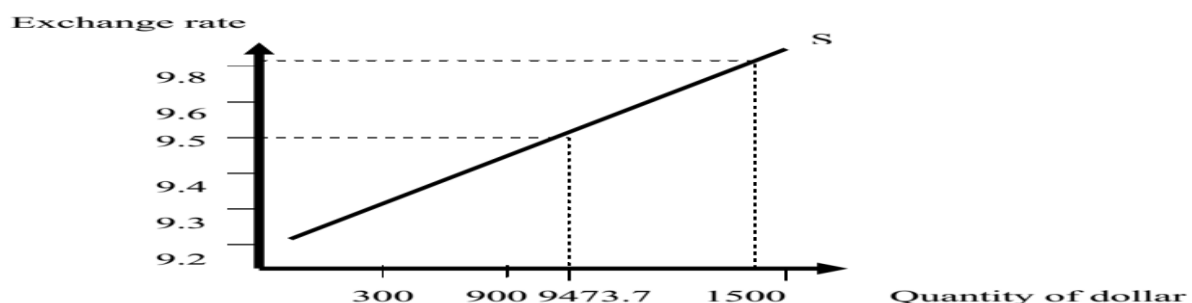
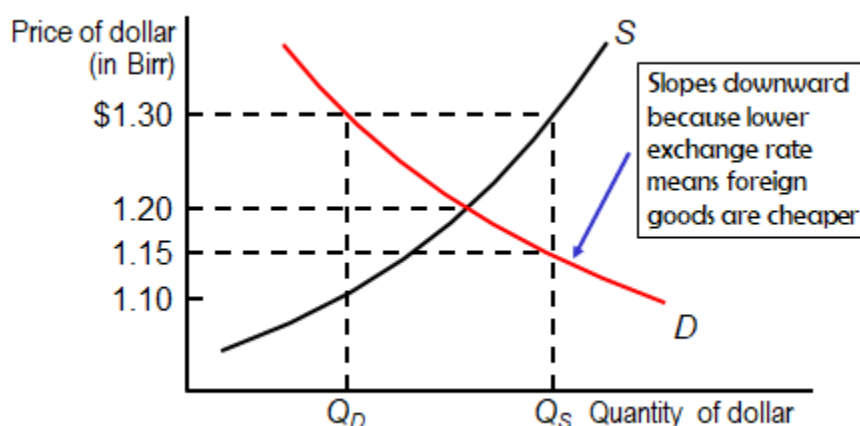


Figure 1.2 The Supply curve of dollar

The Supply of and Demand for Dollar



- Supply of foreign currency is created by: Ethiopian. Exports of goods and services and its capital inflows for Ethiopia.
- Demand for foreign currency is created by: Ethiopian. Imports of goods and services and which is capital outflows for Ethiopia
- Supply of U.S. dollars is created by U.S. imports of goods and services and there are capital outflows from U.S.
- Demand for U.S. dollars is created by U.S. exports of goods and services which results capital inflow to U.S.

Causes of shift in demand for dollar

GDP changes: When a country's income falls, the demand for imports falls. Then demand for foreign currency to buy those imports falls.

Price level changes (inflation): If the U.S. has more inflation than other countries, foreign goods will become cheaper. U.S. demand for foreign currencies will tend to increase, and foreign demand for dollars will tend to decrease.

Interest rate changes: A rise in U.S. interest rates relative to those abroad will increase demand for U.S. assets. The demand for dollars will increase. The forward discount or premium is usually expressed as a percentage of the spot exchange rate. That is,

$$\textbf{Forward discount or Premium} = \frac{F - S}{S} \times 100$$

Where F- is forward exchange rate quotation, S – is the spot exchange rate quotation. The presence of arbitrage ensures that the covered interest parity (CIP) condition holds continuously.

Covered interest rate parity refers to a theoretical condition in which the relationship between interest rates and the **spot and forward currency** values of two countries are in equilibrium. The covered interest rate parity situation means there is no opportunity for arbitrage using forward contracts, which often exists between countries with different interest rates. CIP is the formula used by banks to calculate their exchange quotation and is given by:

$$F = \frac{(r^* - r)}{(1 + r)} + s$$

Where, F is the one-year forward exchange rate quotation in domestic currency per unit of foreign currency, S- is the spot exchange rate quotation in domestic currency, per unit of foreign currency, r is the one year foreign interest rate and r* is the one year domestic interest rate.

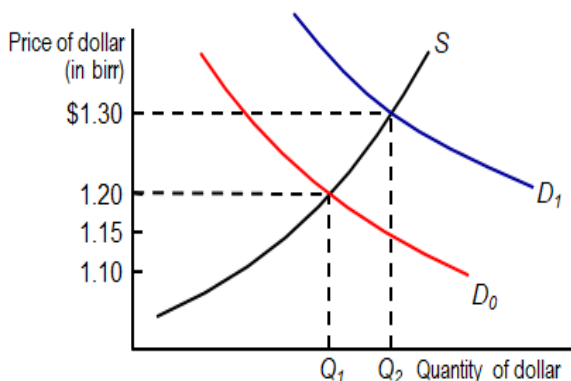


Fig.1.4.1. Demand for Dollar

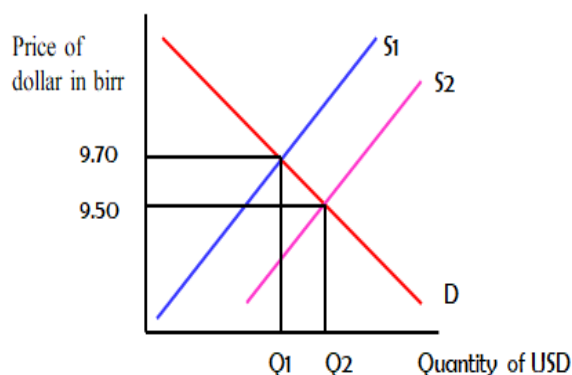


Fig.1.4.2. Supply for Dollar

1.5. Exchange Rate Systems (regimes)

There are three types of exchange rate regimes:

Fixed (or pegged) exchange rate system – the gov't chooses an exchange rate & offers to buy & sell currencies to keep the exchange rate within a narrow band. The “official rate” is called the par value. It is called a **pegged exchange rate**, in which a currency's value is fixed or pegged by a monetary authority against the value of another currency or gold. Under a floating-rate system the fall in the market price (exchange rate) of the currency is **depreciation** and the rise in the market price (exchange rate) of the currency is **appreciation**.

Flexible (floating) exchange rate system – determination of exchange rates is left totally up to the market, and is determined entirely by supply and demand. The major trading countries have been on this system since 1973. Under a fixed-rate system the official lowering of the par value of a currency is **devaluation** and official rising of the par value of a currency is **revaluation**.

Partially flexible (dirty or managed float) exchange rate system– the government sometimes affects the exchange rate & sometimes leaves it to the market.

✚ The effects of changes in exchange rate on Import/ Export

When a country's currency depreciates or is devalued: Foreigners find its exports are cheaper, & the volume of exports rises. Residents find that imports are more expensive, and the volume of import falls. Hence, net exports rise & GDP rises.

When a country's currency appreciates or is revalued (upward): Foreigners pay more for its exports, & the volume of exports falls. Residents pay less for imports, and the volume of imports rises. Hence, net exports fall and GDP falls.

✚ Direct intervention of authorities on foreign exchange rate

If the **exchange rate is floating** no authority intervention is required. Because the market will automatically adjust the exchange rate fluctuation based on the demand and supply forces. Basically, it is based on the trade balance. i.e. if trade balance shows surplus, exchange rate will appreciate. If balance shows deficit, exchange rate will depreciates.

Fixed exchange rate: To maintain a given fixed exchange rate, a country must stand ready to intervene in the foreign exchange market, buying or selling (support or defend) its currency to maintain the official par value. A country can maintain a fixed exchange rate only as long as it has the official reserves (foreign currencies) to maintain this constant rate.

Intervention to control revaluation

Once it runs out of official reserves, it will be unable to intervene, and then must either borrow or devalue its currency.

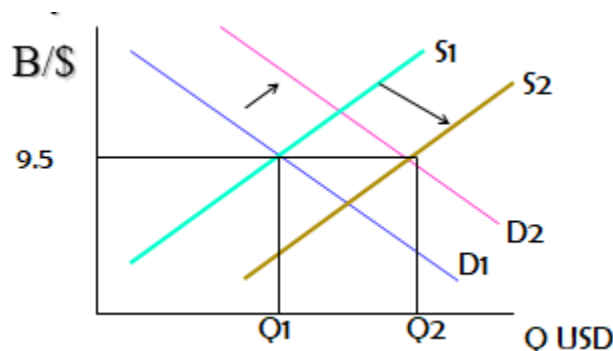


Fig.1.5. Fixed exchange rate: intervention to control revaluation

In the above figure, the increased demand for dollar forced the dollar to be revalued. However, the authority needs to control dollar from revaluation. Thus, intervene in FORX, by selling Q1Q2 amount of dollar to purchase home currency.

This action increases dollar supply from S_1 to S_2 , which eliminates the excess demand for dollar. So that the exchange rate will remain fixed at 9.5B/\$.

Intervention to control devaluation

Increased demand for Ethiopian export will increase supply of dollar, this will in turn rise demand for birr. As result forces the birr to revalue/ dollar to devalue/ shifts supply of USD from S_1 to S_2 . But the authority is not willing to revalue birr. Thus, it will intervene in FORX to collect the excess supply of dollar through selling Birr for dollar. This action raises the foreign reserve and birr in circulation.

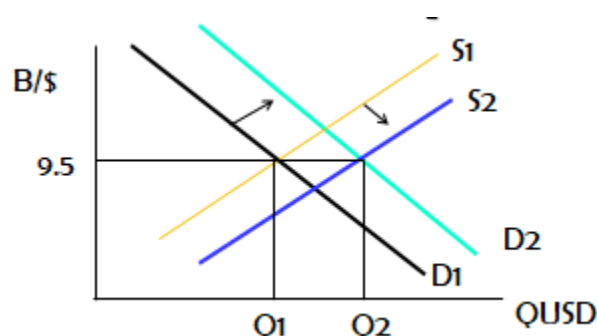


Fig. 1.6. Fixed exchange rate: intervention to control devaluation

To maintain the fixed exchange rate, Q_1Q_2 amount of dollar has to be purchased. This action eliminates the excess demand for birr & maintain the exchange rate fixed at 9.5b/\$.

Transaction in forward exchange market: There are three kinds of players and three kinds of transaction in the forward exchange rate market: Arbitrage: Hedging and Speculation.

A. Arbitrage

Because of the foreign exchange market is so large and decentralized, tiny discrepancies/ difference between exchange rates and cross-rates may briefly arise. Arbitrage is the process of simultaneously buying and selling of currencies to take advantage of such discrepancies. i.e. to make profit.

Importance of arbitrage; it is nearly riskless; it ensures that rates in different locations are essentially the same, and it ensures that rates and cross-rates are consistent. It transfers foreign

exchange from where it is cheap to where it is expensive. It works as stabilizing factor in foreign exchange market.

Weakness: It is possible only when there is **no exchange rate control**. Otherwise, it is impossible.

Types of Arbitrage

We said that arbitrage is exploitation of price differential in the exchange rate. There are two types: financial - center arbitrage and cross-currency arbitrage

Assumption of arbitrage

- ✓ Transaction costs are zero (negligible)
- ✓ There is only single exchange rate quotation

Financial center arbitrage: this ensures birr –dollar exchange rate quoted in New York will be same as that quoted in Addis Ababa is one the financial center. If exchange rate is birr 9.22 /\$ in A.A but birr 9.20 /\$ in NY it would be profitable for banks to buy birr in A.A and sell in NY for a guaranteed profit of 0.02cents such buying of currency from where they are cheap and selling it at higher rate is called **financial center arbitrage**.

Cross–currency arbitrage: this arbitrage involves three currency units. Suppose that the exchange rate of birr is 9.20/\$ and the exchange rate of dollar against Euro is 1.5\$/euro. Currency arbitrage implies that the exchange rate against Euro is 13.8birr/euro = (1.5*9.2) If the actual Birr/Euro =14, then the US dealer wanting birr would do better to first obtain Euro with 13.8Birr/euro(1.5*9.2), then sell birr 14, making 0.2cents per each Euro. If you arbitrage through three exchange rates, this is called **triangular arbitrage**.

The discrepancies b/n exchange rates, is known as the **forward discount or forward premium**. A currency is said to be at a **forward premium** if its forward exchange rates represents an appreciation as compared to the spot rate quotation (positive). A currency is said to be at **forward discount** if its forward exchange rate quotation shows **depreciations (negative)**. The forward discount or premium is usually expressed as a percentage of the spot exchange rate. That is,

$$\text{Forward Discount or Premium} = \frac{F - S}{S} \times 100$$

Where F- is forward exchange rate quotation, S is the spot exchange rate quotation.

The presence of arbitrage ensures that the covered interest parity (CIP) condition holds continuously.

Covered interest rate parity refers to a theoretical condition in which the relationship between **interest rates and the spot and forward currency** values of two countries are in equilibrium.

The covered interest rate parity situation means there is no opportunity for arbitrage using forward contracts, which often exists between countries with different interest rates.

Covered interest parity (CIP) is the formula used by banks to calculate their exchange quotation and is given by;

$$F = \frac{(r^* - r)}{(1 + r)} + S$$

Where, F- is the one year forward exchange rate quotation in domestic currency per unit of foreign currency, S-is the spot exchange rate quotation in domestic currency, per unit of foreign currency, r is the one year foreign interest rate and r* is the one year domestic interest rate.

As an illustrative example of the determination of the forward exchange rate, suppose that the Euro interest rate is 5% and the birr interest rate is 3% and the spot rate birr against euro is birr 10/euro. The one year forward exchange rate can be calculated as:

$$F = \frac{(0.03 - 0.05)}{(1 + 0.05)} + 10 = \frac{-0.02}{1.05} + 10 = 9.81$$

$$\text{Forward discount/premium} = \frac{9.81 - 10}{10} * 100 = -0.019 * 100 = -0.02$$

This shows that the one year forward rate of euro is at an annual forward discount of 2%.

For example birr 10.1/Euro. In this instance an Ethiopian investor with birr 1000 could base on the above example earn 1050 birr at the end of the year(1.05x1000), but by buying Euro spot at birr 10 per euro, and simultaneously selling forward at birr 10.10 at spot exchange rate he would buy 100 Euro (1000/10) and will earn 105 at the end of the year at 5%Euro interest rate (1.05x100=105). Selling 105 at forward rate of 10.1 giving him birr 1060.5 = (105x10.1). This process continues until arbitrage possibility eliminates. With a spot rate of birr10/Euro, only the

forward rate is at 9.81 birr will yield in Ethiopia and in European Union time deposit be identical (since $105 \times 9.81 = 1030$). Only at this forward rate risk-less arbitrage profits to be made.

$$F = \frac{(r^* - r)s + s}{(1 + r)} = \frac{r^* - r}{1} = \frac{F - S}{s}$$

If the country interest rate is higher than the foreign interest rate, then the currency be at forward premium by equivalent percentage. If the domestic interest rate is lower than the foreign interest rate, the currency will be at forward discount by an equivalent percentage. For example, if interest rate on birr 3% & euro interest rate 5% per annual indicates an annual forward discount on euro of 2% => birr will depreciate by nearly 2% against euro.

Other kinds of Trades

Traders may engage in hedging (they may hedge) to reduce or eliminate exposure to exchange risk, by eliminating a net asset or net liability position in the foreign currency.

Forex hedge is a transaction implemented to protect an existing or anticipated position from an unwanted move in exchange rates. Forex hedges are used by a broad range of market participants, including investors, traders and businesses. By using a forex hedge properly, an individual who is long a foreign currency pair or expecting to be in the future via a transaction can be protected from downside risk. Alternatively, a trader or investor who is short a foreign currency pair can protect against upside risk using a forex hedge.

Speculating is the act of taking a long position or a short position in some currency or related asset in hopes of making a short-term profit. It is purely a gamble, & is not motivated by any import/export activity.

B. Hedging

A forward exchange contract is an agreement to exchange one currency for another at some specified future date at an exchange rate set now (the forward exchange rate). The exchange rate that actually eventuates is called **the future spot rate**.

Hedging involves acquiring an asset in the foreign currency to offset a net liability in another, or vice versa. It effectively sets the exchange rate for a future exchange transaction now, removing the exchange rate risk. If 100% of the risk is removed, it is called a **perfect hedge**. For example,

a U.S. firm makes a sale to a company in another country for \$10 million worth of merchandise for delivery in 30 days and payment 60 days thereafter in the foreign currency. Rather than risk the exchange rate changing unfavorably, the U.S. firm enters into a forward exchange contract with a bank, guaranteeing that the bank will exchange the foreign currency for U.S. dollars at the current rate upon receipt of the payment in 90 days. In payment for this contract, the bank receives a forward premium. This is something like buying an insurance policy, guaranteeing the exchange rate at which the exchange will be made.

Suppose US radio importer imports radio from Japan Sony Company. But the importer knows that in 30 days it must pay yen to the Sony Company for shipment of radios arriving then. The importer can sell each radio for \$100 & must pay its supplier yen 9000/radio, its profit depends on the dollar/yen exchange rate. At the current spot exchange rate of \$0.0105/yen. The radio importer firm would pay $\$0.0105/\text{yen} \times 9,000 \text{ yen} = \94.50 per radio; Profit = $\$100 - \$94.50 = \$5.50/\text{radio}$ But the importer may not have the fund to pay to the supplier until the radio arrives & are sold. If on the next 30 days dollar unexpectedly depreciated to \$0.0115/yen. The importer firm will have to pay $\$0.0115/\text{yen} \times 9000 = \$103.50/\text{radio}$. Profit = $\$100 - \$103.50 = -\$3.50$ lose incurred due to appreciation in yen.

To avoid such risk of lose, the importing firm can make 30 days forward exchange deal with a Bank of America. If the Bank of America agrees to sell yen to importer in 30 days at the rate of \$0.0107/yen, then the importer is ensured of paying $\$0.0107/\text{yen} \times 9000 = \96.30 , Profit = $\$100 - \$96.30 = \$3.70$.

C. Speculation

Speculation is a deliberate act of assuming risk to make profit from the fluctuation in exchange rate. The profit from speculation depends on the speculator's expectation about the future. There are **pessimistic/bears/ speculators**: expect exchange rate to decline, thus sell their currency holding to avoid loss. **Optimistic /bulls/ speculator's**: expect exchange rate to increase, they buy foreign currency with a view to selling it when exchange rate increase in future. Speculative transaction has both stabilizing and destabilizing impact on the exchange rate. If a speculator buy a currency when it is cheap and sell when it is dearer, it has a stabilizing effect on exchange rate(Bulls).

If speculators sell when a currency is weak in the expectation that it will be weaker, or buying it when it rises in prices believing that it will be still more stronger /Bears/. Has destabilizing effect on exchange rate

In general, if speculators buy when a currency is weak and sell when it is strong, the speculation is stabilizing. And vice versa. Examples: *A trader purchases a currency—i.e., takes a long position. If the exchange rate moves in favor of that currency viz a viz another currency, the trader sells the currency (“closes out the long position”) at a profit (in the other currency).*

A trader sells a currency that he/she does not own—i.e, takes a short position. The trader takes the proceeds from the sale and holds it in his/her account. If the exchange rate moves lower, the trader buys back the currency at a lower price, and keeps the leftover money (profit).

Example: Suppose that the dollar exchange rate for the British pound is \$2.00. You believe (know?) that the exchange rate in 90 days will be \$1.50. And you offer a forward contract, agreeing to provide £10,000,000 for \$5,000,000 in 90 days. In 90 days, the exchange rate is \$1.50. You buy \$5,000,000 with £7,500,000 and keep £2,500,000 as profit.

Covered and Uncovered Investments

A covered international investment is one for w/c the foreign exchange is fully hedged. An uncovered international investment is one for w/c the foreign exchange is not hedged. The covered interest differential (CD) in favor of a UK investment (“in favor of London”) is:

$$CD = (1 + r_{UK})F/S - (1 + r^*_{US})$$

where F is the forward exchange rate, S is the spot exchange rate, r^*_{UK} and r_{US} are the interest rates in the U.K. and the U.S., respective.

Interest Differentials

The forward premium (or discount, if negative) is the proportionate difference b/n the current forward exchange rate & the current spot value; that is,

$$\text{Forward premium} = (F - S)/S$$

Thus, approximately,

$$CD = \underbrace{F + (r_{UK} - r^*_{US})}_{\text{Covered return, UK} \quad \text{Return, US}}$$

Covered return, UK Return, US

Returns on foreign investments are always the sum of the foreign exchange gain/loss and the local return on the investment.

Covered Interest arbitrage

Covered interest arbitrage is buying a country's currency in the spot market and selling it forward, while making a net profit off: the combination of higher interest rate in the country and any forward premium on its currency.

Covered interest parity

John Maynard Keynes argued that opportunities for arbitrage profits should be self-eliminating. Because the forward exchange rate would adjust so that the covered interest differential returned to zero. After Keynes, we refer to $CD = 0$ as covered interest parity, specifically, Covered interest parity: any interest rate differential between countries should be offset exactly by the forward premium or discount on its exchange rate. That is, the forward premium should be approximately equal to the difference in interest rates. Covered interest parity helps explain differences b/n spot and forward exchange rates.

Evidence on covered interest parity: One study examined CDs b/n short-term financial assets in the U.S. and those in Germany, Japan, and France. For Germany and Japan, the covered interest differential is consistently very close to zero (within the bounds of transactions costs) from about 1985 on. For France this is true from about 1987. Earlier years showed discrepancies explainable by capital controls that limited the ability of investors to move currencies in or out of the countries in question. Thus, covered interest parity seems to hold.

Uncovered Investment

If the future exchange rate is not “guaranteed” by a forward contract, then the investor must make the decision to invest based upon the future expected spot rate, and the expected uncovered interest differential (EUD) is:

$$EUD = (1 + r_{UK})S_e/S - (1 + r^*_{US})$$

If this is positive, then the expected overall return favors investing abroad; if negative, investing at home.

Uncovered interest parity

Uncovered interest rate parity (UIP) theory states that the difference in interest rates between two countries will equal the relative change in currency foreign exchange rates over the same period. It is one form of interest rate parity (IRP) used alongside covered interest rate parity. The market will drive rates until there is no incentive for shifts in investments—when the expected uncovered differential equals zero, at least for the average investor. If true, then $EUD = 0$, called uncovered interest parity. Equivalently, the expected rate of appreciation of the spot exchange rate of a currency should (approximately) equal the difference in interest rates.

Evidence on uncovered interest parity: This is harder to test because one must know what market participants “expected”. Based on survey data, panel studies of the U.S. versus Germany and Japan suggest that market participants often expected large uncovered differentials. This suggests that uncovered interest parity does not hold very well.

Other studies suggest that uncovered interest parity applies roughly, with important deviations. Exchange rate risk may matter—investors may not feel that they will be adequately paid for accepting risk.

Forward predict future spots: Expectations of market participants about future spot prices appear to be biased. Implications:-The market may not be efficient and market participants learn slowly; that is, their expectations will ultimately be unbiased, but until they have fully absorbed all information, they will appear biased.

There may be problems in the forward rate that prevent it from being an unbiased predictor of the future spot rate & the forward rate is not a particularly accurate predictor, either.

Currency Futures: are future contract for currencies that specify the price of exchanging one currency for another at future date. Currency futures are used to hedge the risk of receiving payments in a foreign currency. Currency futures are contracts traded on organized exchanges, like the Chicago Mercantile Exchange or the NY Futures Exchange (NYFE). The futures contract is a standardized contract, and is a tradable security. It is standardized according to amount, terms, and delivery date, and cannot be customized to the specific buyer.

When you enter into a futures contract, the exchange requires you to put up a specific margin (down payment) in cash. Forward contracts may not require this. Profits & losses accrue to you daily with a futures contract:-it is “marked to market daily”& losses may require you to put up more margins. Forwards profits or losses do not generally accrue until the maturity date. Anyone can enter into a futures contract, not just money center banks dealing in large sums of money. Even “small guy” can get into futures.

Currency options

A **currency option** (also known as a forex option) is a contract that gives the buyer the right, but not the obligation, to buy or sell a certain currency at a specified exchange rate on or before a specified date. For this right, a premium is paid to the seller. Currency options are one of the most common ways for corporations, individuals or financial institutions to hedge against adverse movements in exchange rates.

A currency option gives the buyer or holder of the option the right, but not the obligation, to buy (a call option) or sell (a put option) foreign currency at some time in the future at a price set today. The price at which the buyer has the right to buy or sell is called the **strike price or exercise price**. For this right, the buyer pays the seller a premium.

Currency Swaps

A currency swap, sometimes referred to as a cross-currency swap, involves the exchange of interest and sometimes of principal in one currency for the same in another currency. Interest payments are exchanged at fixed dates through the life of the contract. It is considered to be a foreign exchange transaction and is not required by law to be shown on a company's balance sheet.

In a currency swap, two parties agree to exchange flows of d/t currencies during a specified period of time. It is basically a set of spot and forward exchanges packaged together in a single contract. It generates lower transactions costs than an array of equivalent spot and forward contracts, and also may lower risk.

Review Questions

1. The function of the foreign exchange market is to (A) transfer funds from one nation to another, (B) provide short-term credits to finance trade, (C) provide the facilities for hedging, (D) all of the above.
2. If \$3.60 is needed to purchase £2, exchange rate is (A) $\$3.60 = \text{£}2$, (B) $\$1.80 = \text{£}1$, (C) $\text{£}0.50 = \$1$, (D) $\text{£}0.40 = \$1$.
3. When the Ethiopian demand for dollar increase under a flexible exchange rate system, (A) The birr depreciates, (B) the dollar depreciates, (C) the birr appreciates, (D) none of the above
4. Hedging refers to (A) the acceptance of a foreign exchange risk, (B) the covering of a foreign exchange risk, (C) foreign exchange speculation, (D) foreign exchange arbitrage.
5. Covered interest arbitrage involves, (A) the transfer of liquid funds from one monetary center and currency into another to take advantage of higher interest rate in the latter, (B) hedging of currency, (C), earning extra interest in a riskless way. (D) all of the above.
6. What are the determinants for demand for foreign currency?
7. Suppose the dollar exchange rates of the euro and the yen are equally variable. The euro, however, tends to depreciate unexpectedly against the dollar when the return on the rest of your wealth is unexpectedly high, while the yen tends to appreciate unexpectedly in the same circumstances. As a U.S. resident, which currency, the euro or the yen, would you consider riskier?
8. Suppose the one-year forward $\$/\text{€}$ exchange rate is \$1.26 per euro and the spot exchange rate is \$1.2 per euro. What is the forward premium on euro (the forward discount on dollars)? What is the difference between the interest rate on one-year dollar deposits and that on one-year euro deposits (assuming no political risk)?

CHAPTER TWO

THEORY OF EXCHANGE RATE DETERMINATION (MONEY INTEREST RATE & EXCHANGE RATE)

Chapter Objective

- Defining money, functions of money
- Examine demand for and supply of money
- Interest rate, exchange rate, and price level
- Money supply in short-run and long-run its effect on real variables

2. Definition of Money

What is money?

As the word money is used in everyday conversation, it can mean many things, but to economists, it has a very specific meaning.

Money is defined as anything that is generally accepted in payment for goods & services or in the repayment of debts. Defining money merely as currency is too much narrow for economists. Because recently payments are settled not only in paper or coin currency, checks and saving deposits as payment for purchases, these accounts in effect function as money if they can be quickly & easily converted into currency or checking account deposits. Today the credit cards, Smart cards & ATM help liquidating the saving account deposits. So, the term money is sometimes referring to money supply.

2.1. Functions of Money

Money has the following functions

As a medium of exchange: serve as a medium of exchange, a generally Accepted means of payment. Example, consider the barter trade the trade of goods or services for other goods or services.

Unit of account: serve as a widely recognized measure of value. Prices of goods, services, and assets are typically expressed in terms of money. Exchange rates allow us to translate d/t countries 'money prices in to comparable terms. The convention of quoting prices in money

terms simplifies economic calculations by making it easy to compare the prices of different commodities

As store of value: because money can be used to transfer purchasing power from the present into the future, it is also an asset or a store of value and also no one would be willing to accept it in payment if its value in terms of goods & services evaporated immediately.

2.2. Demand for Money

By individuals: what are the factors that determine the amount of money an individual desires to hold? The demand for money by Individuals is based on their demand for an asset on three characteristics:

1. The expected return the asset offers compared with the returns offered by other assets.
2. The riskiness of the asset's expected return,
3. The asset's liquidity.

How the three considerations listed above influence money demand in the economy's households and firms? Let's consider each of them.

Expected return: Currency pays no interest. Checking deposits often do pay some interest, but they offer a rate of return that usually fails to keep pace with the higher return offered by less liquid forms of wealth. When you hold money, you therefore sacrifice the higher interest rate you could earn by holding your wealth in a gov't bond, a large time deposit, or some other relatively non-liquid asset.

Suppose, for example, that the interest rate you could earn from a U.S. Treasury bill is 10% per year. If you use \$10,000 of your wealth to buy a Treasury bill, you will be paid \$11,000= $((1.1 \times 10000))$ at the end of a year. But if you choose instead to keep the \$10,000 as cash in a safe-deposit box, you give up the \$1000 interest you could have earned by buying the Treasury bill. You thus sacrifice a 10% rate of return by holding your \$10,000 as money.

The **expected return theory** states that, other things equal, people prefer assets offering higher expected returns. An increase in the interest rate implies arise in the rate of return on less liquid assets relative to the rate of return on money. Thus, individuals will want to hold more of their wealth in non-money assets that pay the market interest rate & less of their wealth in the form of

money if the interest rate rises. Therefore, all else equal, a rise in the interest rate causes the demand for money to fall.

Riskiness: It is risky to hold money b/c an unexpected increase in the prices of goods & services could reduce the value of your money in terms of the commodities you consume. Since interest-paying assets such as gov't bonds have face values fixed in terms of money, the same unexpected increase in prices would reduce the real value of those assets by the same percentage. Therefore, changes in the risk of holding money need not cause individuals to reduce their demand for money & increase their demand for interest-paying assets.

Liquidity: Households & firms hold money b/c it is the easiest way of financing their everyday purchases. That is, to finance a continuing stream of smaller expenditures at various times & for various amounts, households & firms have to hold some money. An individual's need for liquidity rises when the average daily value of his transactions rises. In general, a rise in the average value of transactions carried out by a household or firm causes its demand for money to rise.

Aggregate Demand for Money

Aggregate demand: The total demand for money by all households & firms in the economy. Aggregate money demand is just the sum of all the economy's individual money demands. Three main factors determine aggregate money demand:

1. **The interest rate:** A rise in the interest rate causes each individual in the economy to reduce her demand for money. All else equal, aggregate money demand therefore falls when the interest rate rises.
2. **The price level:** The economy's price level is the price of a standard reference basket of goods & services in terms of currency. If the price level rises, individual households & firms must spend more money than before to purchase their usual weekly baskets of goods & services. To maintain the same level of liquidity as before the price level increase, they will therefore have to hold more money.
3. **Real national income:** When real national income (GNP) rises, more goods & services are being sold in the economy. This increase in the real value of transactions raises the demand for money, given the price level.

If P is the price level, R is the interest rate, and Y is real GNP, the aggregate demand for money, M^d , can be expressed as

$$M^d = PxL(R, Y)$$

Where the value of $L(R, Y)$ falls when R rises, and rises when Y rises.

If all prices doubled but the interest rate and real income remained unchanged. The money value of each individual's average daily transactions would then simply double, as would the amount of money each wished to hold.

$$\frac{M^d}{P} = L(R, Y) - \text{real money demand}$$

This way of expressing money demand shows that the aggregate demand for liquidity, $L(R, Y)$, is not a demand for a certain number of currency units but is instead a demand to hold a certain amount of purchasing in liquid form. For a given level of real GNP, changes in interest rates cause movements along the $L(R, Y)$ schedule. Changes in real GNP, however, cause the schedule itself to shift.

The downward-sloping real money demand schedule shows that for a given real income level, Y , real money demand rises as the interest rate falls.

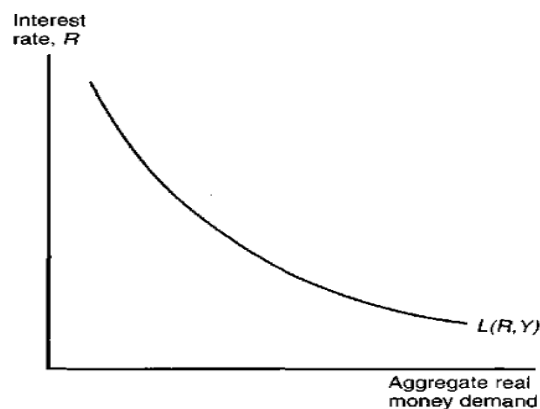


Fig2.1.: Aggregate money demand and interest rate.

An increase in real income from Y^1 to Y^2 raises the demand for real money balances at every level of the interest rate and causes the whole demand schedule to shift upward.

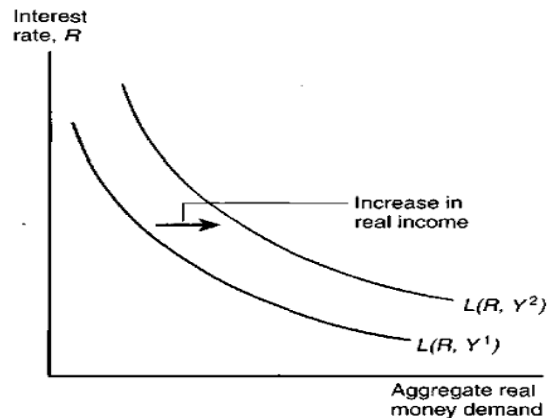


Fig.2.2. Effects of rise in income on aggregate money demand

2.3. Equilibrium in the money market

If M^s is the money supply, the condition for equilibrium in the money is: $M^d = M^s$. After dividing both sides of this equality by the price level, we can express the money market equilibrium condition in terms of aggregate real money demand as: $\frac{M^s}{P} = L(R, Y)$. Given the price level O , and output Y , the equilibrium interest rate is the one at which aggregate real money demand equals the real money supply.

The market always moves toward an interest rate at w/c the real money supply equals aggregate real money demand. If there is initially an excess supply of money, the interest rate falls, & if there is initially an excess demand, it rises.

£ **Interest rate and money supply:** An increase in the money supply lowers the interest rate, while a fall in the money supply raises the interest rate, given the price level and output.

For a given price level, P , and real income level, Y , an increase in the money supply from M^1 to M^2 reduces the interest rate from R^1 (point 1) to R^2 (point 2).

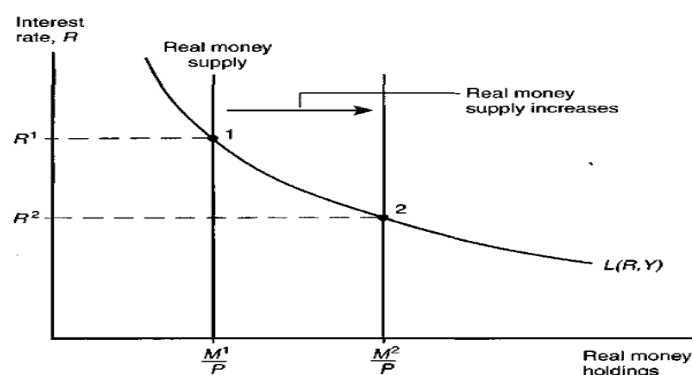


Fig.2.3. Effects of increased money supply on interest rate

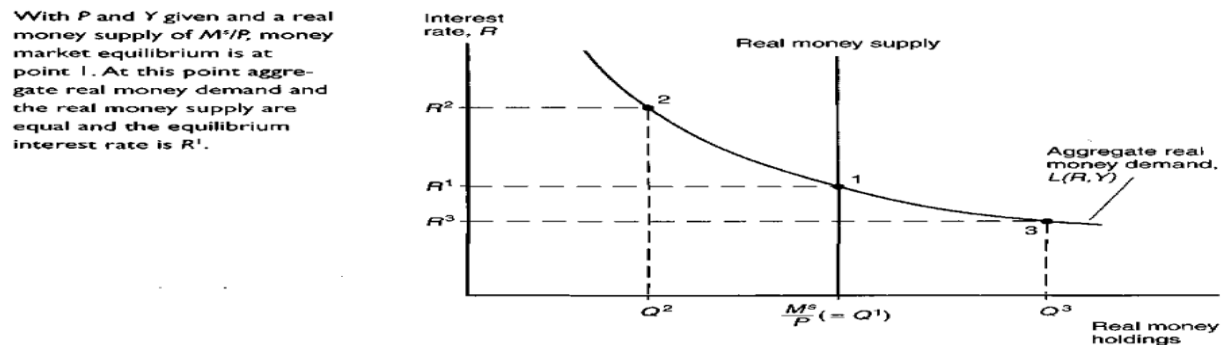


Fig.2.4. Determination of equilibrium interest rate.

£ The money supply and exchange rate in the short run

In previous chapter, we learned about the interest parity condition, which predicts how interest rate movements influence the exchange rate, given expectations about the exchange rate's future level. How shifts in a country's money supply affect the interest rate on non-money assets denominated in its currency, & how monetary changes affect the exchange rate? We will discover that an increase in a country's money supply causes its currency to depreciate in the foreign exchange market, while a reduction in the money supply causes its currency to appreciate.

Linking Money, the Interest Rate, and the Exchange Rate

To analyze the relationship between money & the exchange rate in the short run, let's combine money market and exchange market equilibriums. Let's assume once again that we are looking at the dollar/euro exchange rate, i.e., the price of euros in terms of dollars.

The equilibrium in the foreign exchange market shows how exchange rate is determined given interest rates & expectations about future exchange rates. While equilibrium in the money market reveals how the dollar interest rate is determined in the money market.

£ Combination of equilibrium in two markets, i.e., money market and foreign exchange market.

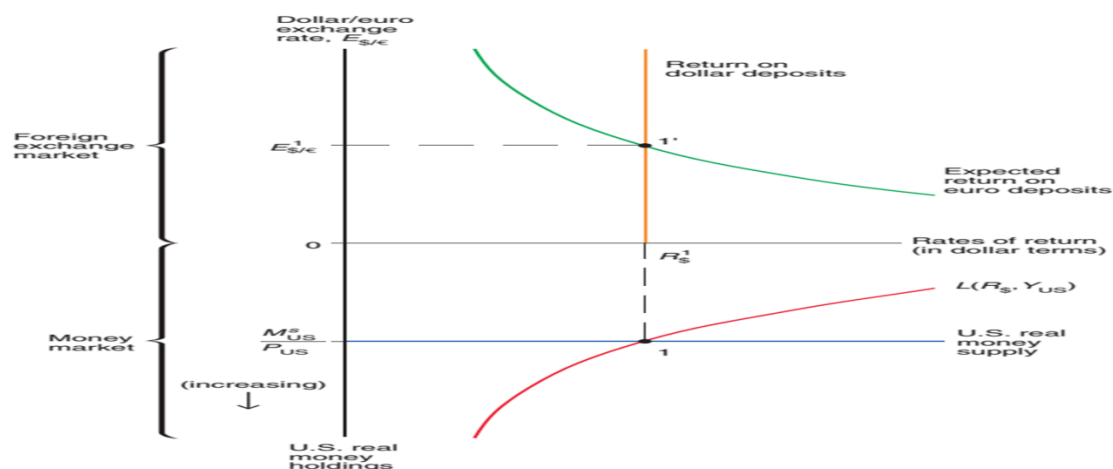


Fig.2.5. Simultaneous Equilibrium in the U.S. Money Market and the Foreign Exchange market

The top figure reveals the equilibrium in foreign exchange market and how exchange rate is determined given interest rate and expectations about future exchange rates. The downward-sloping curve shows the expected return on euro deposits, measured in dollars. It is downward sloping because of the effect of current exchange rate changes on expectations of future depreciation.

Strengthening of dollar today (a fall in $E_{\$/\epsilon}$) relative to its given expected future level makes euro deposits more attractive by leading people to anticipate a sharper dollar depreciation in the future. At the intersection of the two schedules (point 1), the expected rates of return on dollar and euro deposits are equal, and therefore interest parity holds, $E_{\$/\epsilon}$ is the equilibrium exchange rate. While the bottom figure depicts equilibrium in the money market and how the dollar interest rate is determined in the money market.

Money market is at equilibrium at point 1, where the dollar interest rate $R_{\1 induces the demand and supply for real balances are equal. The U.S money market determines the dollar interest rate, which in turn affects the exchange rate that maintains interest parity. The same is true for European money supply.

Given the price levels and national incomes of the two countries equilibrium in national money markets leads to the dollar and euro interest rates $R_{\$}$ and R_{ϵ} . These interest rate feed into the foreign exchange market, where, given expectations about the future dollar/ euro exchange rate, the current rate is determined by the interest parity condition.

Link between Money and Exchange rate

How the dollar/euro exchange rate changes when the federal reserve changes the U.S. money supply? The effect of this change in money supply is summarized in the following figure.

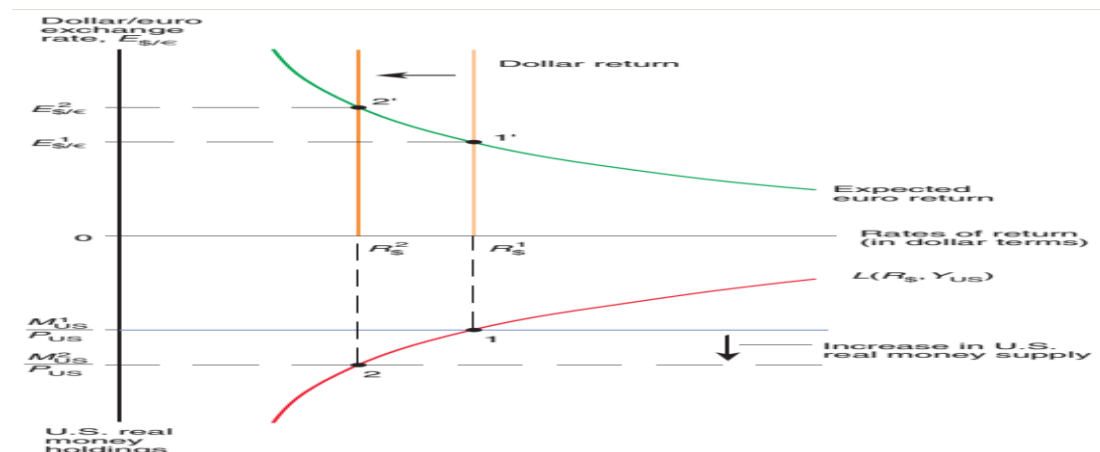


Fig. 2.7. effect on the dollar/euro exchange rate and dollar interest rate of an increase in the U.S. money supply.

At the initial money supply M_{US} , the money market is in equilibrium at point 1 with an interest rate $R_{\$1}$. Given the euro interest rate and the expected future exchange rate, a dollar interest rate of $R_{\$1}$ implies that foreign exchange market equilibrium occurs at point, with an exchange rate equal to $E_{\$/\epsilon}$.

What happens when the Federal Reserves, perhaps fearing the onset of a recession, raises the U.S. money supply to M_{US^2} ? This increase sets in motion the following sequence of events:

- 1) At the initial interest rate, there is an excess supply of money in the U.S. money market , so the dollar interest rate falls to $R_{\2 as the money market reaches its new equilibrium position (point 2).
- 2) Given the initial exchange rate $E_{\$/\epsilon}^1$ and the new lower interest rate on dollars $R_{\2 , the expected return on euro deposits is greater than that on dollar deposits. Holders of dollar deposits therefore try to sell them for euro deposits, which are momentarily more attractive.

- 3) The dollar depreciates to $E^2_{\$/\epsilon}$ as holders of dollar deposits bid for euro deposits. The foreign exchange market is once again in equilibrium at point 2' b/c the exchange rate's move to $E^2_{\$/\epsilon}$ causes a fall in the dollar's expected future depreciation rate sufficient to offset the fall in the dollar interest rate.

All in all an increase in a country's money supply causes its currency to depreciate in the foreign exchange market. And a reduction in a country's money supply causes its currency to appreciate in the foreign exchange market. What do you think the effect of the change in money supply of European on the return on dollar?

£ Money, the Price Level, & the Exchange Rate in the Long Run

How monetary factors affect a country's price level in the long run? Long-run equilibrium is the position an economy would eventually reach if no new economic shocks occurred during the adjustment to full employment. The long-run equilibrium that would occur if prices were perfectly flexible & always adjusted immediately to preserve full employment, we need to examine how such changes shift the economy's long-run equilibrium. Using the theory of aggregate money demand.

If the price and output are fixed in short run, the condition for money market equilibrium, determines the domestic interest rate, R

$$\frac{M^s}{P} = L(R, Y)$$

By rearranging, we can determine the long-run price level, when the interest rate and output are at their long-run level that is at levels consistent with full employment.

$$P = \frac{M^s}{L(R, Y)}$$

This shows how the price level depends on the interest rate, real output, and the domestic money supply. When money market is in equilibrium & all factors of production are fully employed, then the price level will remain steady if the money supply, the aggregate money demand function, and

the long-run values of R and Y remain steady.

The predictions of the above equation for P from the relationship between a country's price level and its money supply, in general is : All else equal, an increase in a country's money supply causes a proportional increase in its price level. For example, if the money supply doubles ($2M_s$) but Y and R do not change, the price level must also double ($2P$) to maintain equilibrium in the money market. That is, demand for real balance is independent of change in money supply that leaves R and Y . If aggregate real money demand is independent of money supply, then money market equilibrium will be maintained is the rise in money supply is totally absorbed by the rise in price level.

£ The Long-Run Effects of Money Supply Changes

We have seen that money supply affects the price level but not interest rate & output, which can't explain how money supply changes affect the price level in the long run. To understand the long-run effects of money supply on the interest rate and output, consider a currency reform, where government redefines the national currency unit. For example, if the government of Ethiopia reformed birr to each new birr equals 1000 old birr, this effect is simply to affect the number of money in circulation and prices but no effect on real variables. An increase in the supply of a country's currency has the same effect in the long run as a currency reform.

That is, doubling of the money supply, for example, has the same long-run effect as a currency reform in which each unit of currency is replaced by two units of "new" currency. If the economy is initially fully employed, every money price in the economy eventually doubles, but real GNP, the interest rate, and all relative prices return to their long-run or full-employment levels.

- *Why change in money supply is similar with a currency reform in its effects on the economy's long-run equilibrium?*

The full-employment output level is determined by the economy's endowments of labor and capital, i.e. long run, real output is independent of money supply.

If money supply and all prices double permanently, there is no reason why people previously willing to exchange \$1 today for \$1.10 a year from now will not be willing to exchange \$2 today for \$2.20 a year from now, so the interest rate will remain at 10% per annum.

Relative prices also remain the same if all money prices double, since relative prices are just ratios of money prices. Similarly, the interest rate is independent of the money supply in the long run. That is, changes in money supply don't change the long-run allocation of resources. but the absolute level of money prices changes.

A permanent increase in the money supply causes a proportional increase in the price level's long-run value. In particular, if the economy is initially at full employment, a permanent increase in the money supply eventually will be followed by a proportional increase in the price level.

£ Money and the Exchange rate in the Long run

The domestic currency price of foreign currency is one of the many prices in the economy that rise in the long run after a permanent increase in the money supply. Suppose, for example, that the U.S. government replaced every pair of "old" dollars with one "new" dollar. Then if the dollar/euro exchange rate had been 1.20 old dollars per euro before the reform, it would change to 0.60 new dollars per euro immediately after the reform. That is, halving of the U.S. money supply would eventually lead the dollar to appreciate from an exchange rate of 1.20 dollars/euro to one of 0.60 dollars/euro.

Since the dollar prices of all U.S. goods & services would also decrease by half, this 50 % appreciation of the dollar leaves the relative prices of all U.S. & foreign goods & services unchanged.

Generally, all else equal, a permanent increase in a country's money supply causes a proportional long-run depreciation of its currency against foreign currencies. Similarly, a permanent decrease in a country's money supply causes a proportional long-run appreciation of its currency against foreign currencies.

£ Exchange rate Overshooting

In its initial depreciation after a money supply rise, the exchange rate jumps from $E^1_{\$/\epsilon}$ up to $E^2_{\$/\epsilon}$, a depreciation greater than its long-run depreciation from $E^1_{\$/\epsilon}$ to $E^3_{\$/\epsilon}$ as indicated in the figure below. The **exchange rate is said to overshoot** when its immediate response to a disturbance is greater than its long-run response.

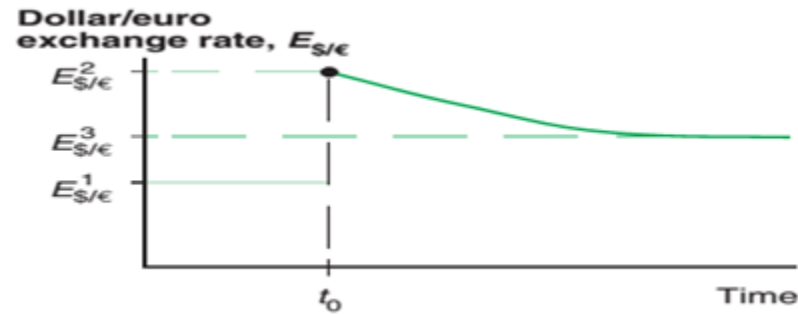


Fig. 2.8. Exchange rate overshooting

Exchange rate overshooting is an important phenomenon because it helps explain why exchange rates move so sharply from day to day. Investors *can* react rationally to news in such a way as to drive the exchange rate *past* what they know to be its ultimate long-run equilibrium rate, and then slowly back to that rate later on. That is, in the short run, the exchange rate actually overshoots its long-run value and then reverts back toward it.

2.4. Price Levels and Exchange Rates in the Long run

- *What economic forces lie behind such dramatic long-term movements in exchange rates?*

We have seen that exchange rates are determined by interest rates and expectations about the future, which are, in turn, influenced by conditions in national money markets. To understand fully long-term exchange rate movements, we have to extend our model in two directions.

First, we must complete our account of the linkages among monetary policies, inflation, interest rates, and exchange rates.

Second, we must examine factors other than money supplies & demands, for example, demand shifts in markets for goods and services—that also can have sustained effects on exchange rates.

2.5. Purchasing Power Parity (PPP)

The **theory of purchasing power parity** states that the exchange rate between two countries' currencies equals the ratio of the countries' price levels. The domestic purchasing power of a country's currency is reflected in the country's price level, the money price of a reference basket of goods and services. The PPP theory therefore predicts that a fall in a currency's domestic purchasing power (as indicated by an increase in the domestic price level) will be associated with a proportional currency depreciation in the foreign exchange market. Symmetrically, PPP predicts that an increase in the currency's domestic purchasing power will be associated with a proportional currency appreciation. Symbolically, PPP can be expressed as :

$$E_{\$/\epsilon} = P_{US} / P_E$$

Where $E_{\$/\epsilon}$ -exchange rate of dollar per euro, P_{US} -price of U.S., P_E -price of European.

Example, if the reference commodity basket costs \$200 in U.S and €160 in Europe. Thus, PPP predicts as $\$/\epsilon$ exchange rate = $\$200/\epsilon160 = \$1.25/\epsilon$. If the U.S price level is tripled, so would the dollar price of a euro. $PPP = E_{\$/\epsilon} = 3 \times 1.25$.

The dollar price of the reference basket when purchased in Europe ($P_{US} = E_{\$/\epsilon} \times P_E$). This is true if the PPP holds. PPP thus, asserts that all countries price levels are equal when measured in terms of the same currencies. Therefore, PPP holds when, at going exchange rate every currency's domestic purchasing power is always the same as its foreign purchasing power.

£ The PPP theory and the Law of One Price

To understand the market forces that might give rise to the results predicted by the purchasing power parity theory, we discuss first a related but distinct proposition known as the **Law of one price**.

Law of one price: states that in competitive markets, free of transportation costs & official barriers to trade (such as tariffs), identical goods sold in d/t countries must sell for the same price when their prices are expressed in terms of the same currency. When trade is open & costless, identical goods must trade at the same relative prices regardless of where they are sold. For

example, suppose that the dollar/pound exchange rate is \$1.50 and the price of sweater is \$45 in New York while the same sweater costs £30 in London. The dollar price of the sweater when sold in London is then $(\$1.50/\text{£ pound}) \times (\text{£}30/\text{sweater}) = \$45/\text{sweater}$, the same as its price in New York.

To see whether the law of one price holds or not, let us further consider the effect of change in exchange rate on the relative prices of the sweater in both markets. If pound depreciated from \$1.50/£ to \$1.45/£, now a sweater costs \$43.50 in London ($=\$1.45/\text{£} \times \text{£}30$). If the same sweater were selling for \$45 in New York, U.S. importers & British exporters would have an incentive to buy sweaters in London & ship them to N.Y, pushing the London price up and the N.Y price down until prices were equal in the two locations and vice versa.

From the above example we can see that the law of one price provides link between the domestic prices of goods and exchange rates.

The law of one price can formally be stated as follows; Let P^i_{US} be the dollar price of good i when sold in the U.S., P^i_E the corresponding euro price in Europe. Then the law of one price implies that the dollar price of good i , is the same wherever it is sold, $P^i_{US} = E_{\$/\text{€}} \times P^i_E$. The dollar/euro exchange rate is the ratio of the price of good i , in U.S. and European money prices is $E_{\$/\text{€}} = P^i_{US} / P^i_E$, law of one price looks like the PPP. However, there is clear difference

between PPP and law of one price. These are:

1. The law of one price applies to individual commodity, while PPP applies to the general price level, which is composite of the prices of all the commodities that enter into the reference baskets.
2. If the law of one price holds for every commodities PPP must automatically holds as long as the reference baskets used to reckon different countries price level are the same.

The proponents of PPP argue that its validity as a long run theory does not require the law of one price to hold exactly. Example, if the goods & services become temporarily expensive in one country than in others, the demand for its currency & its products fall pushing exchange rate (depreciation) & domestic prices back in line with PPP. The opposite situation of relatively

cheap domestic products leads, to currency appreciation and price level inflation. Thus, PPP, asserts that even when the law of one price is not literally true, the economic forces behind it will help eventually to equalize a currency's purchasing power in all countries.

£ Absolute PPP and Relative PPP

The statement that exchange rates equal relative prices level in (the previous equation) is sometimes referred to as absolute PPP. Absolute PPP implies relative PPP, Which states that the % change in the exchange rate b/n two currencies over any period equals the difference b/n the % changes in national price levels.

Relative PPP thus translates absolute PPP from a statement about price & exchange rate levels into one about price and exchange rate changes. It asserts that prices and exchange rates change in a way that preserves the ratio of each currency's domestic & foreign purchasing powers. If for example, the U.S. price level rises by 10 % over a year while price level of Europe's rises by only 5%, the relative PPP predicts a 5 % depreciation of the dollar against the euro.

The dollar's 5% depreciation against the euro just cancels the 5% by which U.S. inflation exceeds European inflation, leaving the relative domestic & foreign purchasing powers of both currencies unchanged. More formally, relative PPP between the United States & Europe would be written as:

$$(E_{\$/\epsilon,t} - E_{\$/\epsilon,t-1}) / (E_{\$/\epsilon,t-1}) = \pi_{US,t} - \pi_{US,t-1}$$

Where π_t denotes an inflation rate (that is $\pi_t = (P_t - P_{t-1})/P_{t-1}$, the percentage change in a price level between dates t and t-1.

Unlike absolute PPP, relative PPP can be defined only with respect to the interval over which price levels and the exchange rate change. The notion of relative PPP thus convenient when we have to rely on gov't price level statistics to evaluate PPP.

As it makes sense to compare % exchange rate changes to inflation differences, when countries base their price level estimates on product baskets that differ in coverage & composition.

Relative PPP is important also b/c it may be valid even when absolute PPP is not. if the factors causing deviations from absolute PPP are more or less stable over time, % changes in relative price levels can still approximate % changes in exchange rates.

2.5.1. A Long-run Exchange Rate Model Based on PPP

The theory of PPP leads to a useful theory of how exchange rates & monetary factors interact in the long run. It is known as the monetary approach to the exchange rate.

The monetary approach is considered as a long-run theory b/c it does not allow for the price rigidities rather it assumes that as if prices can adjust right away to maintain full employment as well as PPP. This implies that in the long run prices are perfectly flexible to equilibrate the variability in output & factor market prices. To develop the **monetary approach's** predictions for the \$/euro exchange rate, let us assume in the long run, the foreign exchange market sets the rate so that PPP holds: $E_{\$/\epsilon} = P_{US}/P_E$.

This equation would hold if there are no market rigidities to prevent the exchange rate & other prices from adjusting immediately to levels consistent with full employment.

At the beginning of this unit we expressed domestic price levels in terms of domestic money demands and supplies. For instance in the United States, $P_{US} = M^s_{US}/L(R_{\$}, Y_{US})$, while in Europe, $P_E = M^s_E/L(R_{\$}, Y_E)$. The money supply M^s and aggregate M^d decreases when the interest rate rises and increases when real output rises.

According to the statement of PPP, the dollar price of a euro is simply the dollar price of U.S. output divided by the euro price of European output. These two price levels, in turn, are determined completely by the supply and demand for each currency area's money.

The monetary approach therefore makes the general prediction that the exchange rate, w/c is the relative price of American & European money, is fully determined in the long run by the relative supplies of those monies and the relative real demands for them. Shifts in interest rates & output levels affect the exchange rate only through their influences on money demand.

In addition, the monetary approach makes a number of specific predictions about the long-run effects on the exchange rate of changes in money supplies, interest rates, and output levels:

Money supplies: other things equal, a permanent rise in the U.S. money supply M^s_{US} causes a proportional increase in the long-run U.S. price level P_{US} . If the U.S. money supply rises by 10 percent, the price in U.S. and \$/€ both eventually rise by 10 percent as well. Thus, an increase in the U.S. money supply causes a proportional long-run depreciation of the dollar against the euro. Conversely, permanent increase in the European money supply causes a proportional increase in the long-run European price level. Under PPP, this price level rise implies a proportional long-run appreciation of the dollar against the euro (which is the same as a proportional depreciation of the euro against the dollar).

Interest Rate: a rise in the interest rate $R_{\$}$ in dollar-denominated assets lowers real U.S. money demand $L(R_{\$}, Y_{US})$. The long-run U.S. price level rises, and under PPP the dollar must depreciate against the euro in proportion to this U.S. price level increase. A rise in the interest rate $R_{\$}$ on euro-denominated assets has the reverse long-run exchange rate effect. Because real European money demand falls, Europe's price level rises. i.e. PPP, reveals dollar appreciate against the euro in proportion to increase Europe's price level.

Output levels: A rise in U.S. output raises real U.S. money demand leading to a fall in the long-run U.S. price level. Under PPP, shows appreciation of the dollar against the euro. Similarly, a rise in European output raises & causes a fall in Europe's long-run price level. PPP predicts the dollar depreciate against the euro.

According to the monetary approach, the U.S. price level drops immediately to bring about a market-clearing increase in the supply of real balances. PPP implies that this instantaneous American price deflation is accompanied by an instantaneous dollar appreciation on the foreign exchanges. The monetary approach leads to the long-run foreign exchange value of a country's currency moves in proportion to its money supply. The theory also rises what seems to be a paradox. We always found that a currency appreciates when the interest rate it offers rises relative to foreign interest rates.

How a rise in a country's interest rate depreciates its currency by lowering the real demand for its money?

2.6. Measurement Problems with PPP

Many researchers have tried to test whether PPP can be used to predict exchange rate or not. Their results are summarized as follows. PPP performs better for countries that are geographically close to one another and where trade linkages are high. Exchange rates have been much volatile than the corresponding national prices level. This is against the PPP hypothesis in which exchange rates are only expected to be as volatile as relative price. PPP holds better in the long-run than in the short-run. The currencies of countries with very high inflation rates relative to their trading partners, most likely would experience depreciation reflecting their high inflation rate. This suggests that PPP is the dominant force in determining their exchange rate. Overall , PPP holds better for traded goods than non-traded goods.

1. Transport costs & restrictions on trade certainly do exist. These trade barriers may be high enough to prevent some goods & services from being traded b/n countries. Moreover some goods are non-traded
2. Existence of Imperfect competition: in goods markets may interact with transport costs & other trade barriers to weaken further the link b/n the prices of similar goods sold in d/t countries.
3. Because the inflation data reported in different countries are based on different commodity baskets, there is no reason for exchange rate changes to offset official measures of inflation differences, even when there are no barriers to trade and all products are tradable.
4. Difference between Capital and Goods Market: Purchasing power parity is based on the concept of goods arbitrage and has nothing to say about the role of capital movement.

According to Rudiger D. (1976) in a world where capital market are highly integrated & goods markets exhibit slow price adjustment. There can be substantial (significant) & prolonged deviation of the exchange rate from PPP.

That is, in short-run good price in both home and foreign country can be considered as fixed ,while the exchange rate adjusts quickly to new information and change in economic policy.

This is the case being exchange rate changes represent deviation from PPP which can be quite substantial and prolonged(extended).

Productivity Differentials: when prices of similar basket of both traded & non-traded goods are converted into a common currency, the aggregate price indices tend to be higher in rich countries than in poor countries. The overall higher price in rich countries is mainly due to the fact that non-tradable goods prices are higher in developed than in developing countries.

The **lower relative price of non-tradable in poor countries** is due to their lower labor productivity b/n developing & developed countries. In other words higher price of non-traded goods in developed countries is mainly due to their higher labor productivity in the traded sector as compared to developing countries.

The generalized version of PPP

One of the major problems with PPP is that it is supposed to hold for all types of goods. However, a more generalized version of PPP provides some useful insights & makes distinction among goods traded. According to general version of PPP goods can be categorized into traded goods and non-traded goods.

Traded goods:- These are goods w/c are susceptible(easily) to international competition. Here belongs most manufacturing goods like Automobile, Electronics products and fuels and the like.

Non traded goods:- are those that cannot be traded internationally at a profit. Their price will not be affected by the international competition. These includes different goods and services like, Hair cut (hair dressing), Restaurant food service, Houses, E.t.c.

The distinction b/n them is due to the fact that the price of traded goods tends to be in line with the international competition, while the price of non-traded goods will be determined predominantly by domestic supply & demand considerations. For instance if a car costs 15,000 £ in the UK & \$ 30,000 in US arbitrage will tend to keep the pound-dollar rates at 2 \$/£. However, arbitrage forces do not play a role in the case of house trade. Similarly, if a hair-cut cost birr 10 in Ethiopia but \$10 in US and the exchange rate is birr 10/\$. No one in the U.S will travel to Ethiopia for a haircut knowing that they can save \$9 because of the time and transport costs involved.

When aggregate price indice is determined both by tradable and non-tradable are considered. Assuming that PPP holds for tradable we have the following locations.

$$P_T = SP_{T*}.$$

Where P_T - price of traded goods in the domestic country measured in terms of the domestic currency, P_T^* the price of traded goods in the foreign country measured in-terms of foreign currency, S - the exchange rate defined as domestic currency units per unit of foreign currency.

The aggregate price index (PI) for the domestic economy is made up of a weighted average of the price of both tradable (P_T) and non-tradable goods (P_N) priced in the domestic currency. We have,

$$P_I = \alpha P_N + (1 - \alpha) P_T \dots \dots (1)$$

Where, α is the proportion of non-traded goods in the domestic price index.

$$P_{I*} = \beta P_{N*} + (1 - \beta) P_T^* \dots \dots (2)$$

Where, β is the proportion of non-traded goods in the foreign price index. Dividing equation (1) and (2) we obtain

$$\frac{P_I}{P_{I*}} = \frac{\alpha P_N + (1 - \alpha) P_T}{\beta P_{N*} + (1 - \beta) P_T^*}$$

If we divide the numerator by P_T and the denominator by $S P_T^*$, which because of the assumption of PPP for tradable goods are equivalent expression, we obtain;

$$\frac{P_I}{P_{I*}} = s \left[\frac{\alpha \left(\frac{P_N}{P_T} \right) + (1 - \alpha)}{\beta (P_{N*}/P_T^*) + (1 - \beta)} \right] \dots \dots (3)$$

This can be rearranged , to give the solution for the exchange rate as

$$S = \frac{P_I}{P_{I*}} \left[\frac{\beta (P_{N*}/P_T^*) + (1 - \beta)}{\alpha \left(\frac{P_N}{P_T} \right) + (1 - \alpha)} \right] \dots \dots \dots (4)$$

The above equation is an important modification to the initial PPP equation. This is because PPP no longer necessary holds in terms of aggregate price indices due to the terms on the right hand side. Furthermore, the equation suggests that the relative price of non-tradable relative to tradable will influence the exchange rate. Testing the PPP using price indices on tradable goods

prices is likely to lead to better results than when using aggregate price indices made up of both types of goods.

Review Questions

1. In short run, a permanent increase in the domestic money supply causes?
 - A. An upward shift in the demand curve, which is greater than that caused by an equal but transitory increases
 - B. A downward shift in aggregate demand curve, which is greater than that caused by an equal but transitory increase.
 - C. An upward shift in the aggregate demand curve, which is smaller than that caused by an equal but transitory increase.
 - D. A downward shift in the aggregate demand curve, which is smaller than that caused by an equal but transitory increase.
 - E. An upward shift in the AA curve, which is greater than that caused by an equal but transitory increase.
2. Explain how an increase money supply affect exchange rate and price level of that a given nation in the long-run?
3. What are the principal functions of foreign exchange market?
4. If the exchange rate between the birr and the dollar changes from 52 birr =\$1 to 63 birr =\$1, what does this mean for the birr? And what does for dollar?
5. Explain the exchange rate overshooting of the dollar that is likely to result from reduction in the rate of interest in the U.S. relative to that in the U.K?

CHAPTER THREE

BALANCE OF PAYMENT

Chapter Objective

- Define balance of payment, components of BOP,
- Balance of payment disequilibrium
- Link between exchange rate and balance of payment
- Approaches to balance of payment, monetary approach, absorption approach, elasticity approach

3.1. Definition of Balance of payments(BOP)

What is Balance of Payment?

The BOP is the statistical record of a country's international transactions over a certain period of time presented in the form of double-entry bookkeeping.

N.B. when we say “a country's balance of payments” we are referring to the transactions of its citizens & gov't.

Usually reporting period for all the statistics included in the account is a year. However, some of the statistics are published more regularly on monthly or quarterly basis. It reveals how many goods & services the country has been exporting & importing & whether the country has been borrowing or lending money to the rest of the world. For the purpose of BOP reporting, a multinationals are treated as being residents in the country in w/c they are located even if their share are actually owned by foreign residents. Another distinction regarding the treatment of international organizations such as, the International Monetary Fund, the World Bank, United Nations & the like, these institutions are treated as being foreign residents even though they may actually be located in the reporting country. Tourists are regarded as being foreign residents if they are in the reporting country for at less than a year.

£ Why balance of payment (BOP)?

The BOP serves a very useful purpose as it yields necessary information for the future monetary policy, fiscal policy and foreign trade policies formulation.

The major uses of the BOP are as follows

1. It provides extremely useful data for economic analysis of country's weakness and strength as a partner in international trade. The economic position of a country can be determined. If BOP shows improvement or deterioration, necessary correction measures may be adopted.
2. It reveals the changes in the composition and magnitude of foreign capital movements. The changes that are deterrent to economic well-being of the country and lead to deterioration in the BOP of the country warrant necessary action by the government.
3. It provides indications of future repercussions (consequence) of a country's past trade performance. If BOP shows a continuous and large deficits over time, it shows growing international indebtedness which may lead to financial bankruptcy, persistent large scale deficit in BOP if its magnitude exceeds the absorption capacity of the country, causes inflation
4. Detailed BOP account reveals weak and strong points in the country's foreign trade relations and thereby forewarns the government for necessary corrective measures especially against the weak points.

£ Collection, Reporting of the BOP

The BOP statistics record all of the transaction between domestic and foreign residents, they purchases or sales of goods, services or of financial assets such as bonds, equities and banking transactions. Reporting figures are usually in terms of the domestic currency of the reporting country. The authorities collect their information from the custom authorities, survey of tourist numbers and expenditure, and data on capital in-flow and outflow is obtained from banks, pension funds, multinationals and investments Agency. The responses from different sources are compiled by government statistical agencies.

3.2. Balance of Payment Accounting

The BOP accounts are those that record all transactions b/n the residents of a country and residents of all foreign nations. In an accounting sense they always balance. This is because they are based up on the principle of double-entry book-keeping. Each transaction b/n domestic & foreign residents has two sides: a receipt & payment, where, both sides are recorded in the BOP statistics. Each receipt of currency from residents of the rest of the world is recorded as a credit

item (carries a plus in the account) while each payment to residents of the rest of the world is recorded as a debt item (carries a minus in the accounts).

3.3. The Components of BOPs

The following were the elements (components) of balance of payments. It composed of the following accounts; the current account, capital account, the balance for official financing (the official reserves account) and statistical discrepancy or difference.

Current account refers to income flows. Includes all imports and exports of goods and services (invisible trade) and unilateral transfers of foreign aid. If the debits exceed the credits, then a country is running a trade deficit. If the credits exceed the debits, then a country is running a trade surplus. It is thought that the CA responds to changes in income and the exchange rate.

Capital account (K): The capital account records change in the assets and liabilities. The capital account measures the difference between domestic sales of assets to foreigners and domestic purchases of foreign assets.

The Official Reserve Account: It is a subdivision of the capital account, is the foreign currency and securities held by government, usually by its central bank, and is used to BOP from year to year. Official reserves assets include gold, foreign currencies, special drawing rights (SDRs), and reserve positions in the IMF. The official reserves increases when there is a trade surplus and decreases when there is a deficit. The balance for official financing shows the net increase or decrease in a country's holdings of foreign currency reserves: A decrease in the official reserves is reported as a credit item (+) and an increase is reported as a debit item (-).

Statistical Discrepancy: There's going to be some omissions and misreported transactions, so we use a "plug" figure to get things to balance.

The BOP must always balance since the accounts are constructed such that this must be true by definition. However, there can be measurement error & unreported borrowing from abroad & there illegal activities. The discrepancy represents a combination of unrecorded current & capital account transactions. This requires the inclusion of what is referred to as a balancing item, to ensure the accounts balance in practice.

3.4. The Trade balance

The trade balance some time referred to as the visible balance because it represents the difference between receipts from export of goods and expenditure on imports of goods which can be visibly seen crossing frontiers.

The receipts for exports are recorded as a credit in the BOP, while the payment for import is recorded as a debit. When the trade balance is in surplus this meant that a country has earned more from its exports of goods than in has paid for its imports of goods.

£ THE CURRENT ACCOUNT BALANCE

The CA balance is the sum of visible trade balance & the invisible balance. The invisible balance shows the difference between revenue received from export of services & payment made for imports of services such as Shopping, Tourism, Insurance and Banking and Transport. In addition receipts & payments of interest, dividends and profits are recorded in the invisible balance as they represent the rewards for investment in overseas companies, bonds and equities. Payment represents the reward to foreign residents for their investment in the domestic economy.

£ THE CAPITAL ACCOUNT BALANCE

The capital account records transactions concerning the movement of financial capital into and out of the country. Capital comes into the country by borrowing, increases in foreign assets in the nation and investment in the country by foreigners. These items are referred to as capital inflows and are recorded as credit items in the BOP. Capital Inflow-lead to payments from foreigners.

Capital outflows are a decrease in the country's holding of foreign assets or increase in liabilities of to foreigners. Usually capital outflows are recorded as debit in the BOP- it presents some confusion to many readers. Capital outflow- lead to payments to foreigners, assets such as stock, bonds, treasury bills.

The easiest way to minimize this problem is that to think of investment by foreigners as export of equity or bonds & sales of foreign investment as an export of those investments to foreigners.

Financial inflows: can take either of two forms: an increase in foreign assets in the nation or a reduction in the nation's assets abroad. For example, when a U.K. resident purchases Ethiopian

stock, foreign assets in Ethiopia increase. This is a capital inflow to Ethiopia and is recorded as a credit in Ethiopia's BOP because it involves the receipt of a payment from a foreigner.

A capital inflow can also take the form of a reduction in the nation's assets abroad. For example, when an Ethiopian sells a foreign stock, Ethiopian assets abroad decrease. This is a capital inflow to Ethiopia (reversing the capital outflow that occurred when the Ethiopian purchased the foreign stock) and is recorded as a credit in Ethiopian BOP because it too involves the receipt of a payment from foreigners.

Financial Outflows: can take the form of either an increase in the nation's assets abroad or a reduction in foreign assets in the nation because both involve a payment to foreigners. For example, the purchase of a U.K. treasury bill by an Ethiopians increases Ethiopia's assets abroad and is a debit because it involves a payment to foreigners.

Similarly, the sale of subsidiary by a Turkey firm reduces foreign assets in Ethiopia and is also a debit because it involves a payment to foreigners.

Table.3.1. Annual balance of payment (BOP): Current Account Balance

Current Account Balance	
1. Export of goods	+150
2. Import of goods	- 200
3. Trade balance	- 50 (1+2)
4. Export of services	+ 120
5. Import of services	- 160
6. Interest, profit and Dividend received	+ 10
7. Interest, profit and dividend paid	-10
8. Unilateral receipts	+30
9. Unilateral payment	-20
10. Current Account balance	-70 (sum 3-9)

Table.3.2. Annual Balance of payment (BOP)

Capital Account Balance	
11. Investment abroad	-30
12. Short –term lending	-60
13. Medium and long –term lending	-80
14. Repayment of borrowing than Row	-70
15. Inward foreign investment	+170
16. Short-term borrowing	+40
17. Medium and long-term borrowing	+30
18. Repayment on loans received from Rest of the world	+50
19. Capital Account Balance	+50 (sum (11-18))
20. Statistical Error	+5 (zero minus (10+19+24))
21. Official settlement balance	-15 (10 + 19+ 20)
22. Change in reserves	rise (-) fall (+) -10
23. IMF borrowing	from (+) repayment to (-) -5
24. Official financial balance	+15 (22+23)

RECORD OF TRANSACTIONS IN THE BOP

To understand exactly why the sum of credits and debits in the BOP should sum to zero we consider some examples of economic transections between domestic and foreign countries There are basically five types of such transactions that can take place. These are

I . Exchange of goods/services in return for a financial asset.

- II. An exchange of goods/services in return for other goods/ services
- III. An exchange of a financial item in return for a financial item
- IV. A Transfer of goods or services with no corresponding transaction (aid (food or military)
- V. A unilateral transfer of financial asset with no corresponding transaction.

Example 1 Boeing of the United States exports a \$100 million aircraft to the United Kingdom which is paid for by British Airways debiting its US bank deposit account by a like amount.

US balance of payments		UK balance of payments	
current account		current account	
Exports of goods	+\$100m	Import of goods	−£50m
Capital account		Capital account	
Reduced US bank liabilities to UK residents	−\$100m	Reduction in US bank deposit assets	+£50m

Example 2 The US exports \$2,000 of goods to the UK in exchange for \$2,000 of services.

US balance of payments		UK balance of payments	
current account		current account	
Merchandise exports	\$2,000	Exports of services	£1,000
Imports of services	−\$2,000	Imports of goods	−£1,000

Example 3 A US investor decides to buy £500 of UK Treasury bonds and to pay for them by debiting his US bank account and crediting the account of the UK Treasury held in New York.

US balance of payments		UK balance of payments	
capital account		capital account	
Increase in UK treasury bond holdings	−\$1000	Increased bond liabilities to US residents	£500
Increased in US bank liabilities	+\$1000	Increased US bank deposit	−£500

Example 4 The US makes a gift of £1 million of goods to a UK charitable organization.

US current account		UK current account	
Exports	\$2m	Imports	-£1m
Unilateral payment	-\$2m	Unilateral receipt	+£1m

Example 5 The US pays interest, profits and dividends to UK investors of \$100 million by debiting US bank accounts which are then credited to UK residents' bank accounts held in the US.

US current account		UK current account	
Interest, profits, dividends paid	-\$100m	Interest, profits, dividends received	+£50m
US capital account		UK capital account	
Increased US bank liabilities	+\$100m	Increase in US bank deposits	-£50m

3.5. BOP Equilibrium and Disequilibrium (Surplus or Deficit)

BOP always balances, since each credit in the account has a corresponding debit. However, this does not mean that each of the individual accounts that make-up the BOP is necessarily in balance. For instance the current account can be in surplus while the capital account is in deficit.

£ Surplus Vs. deficit in BOP

The CA surplus and deficit is important economic indicator because a surplus means the country is earning more than its spending against the rest of the world (ROW), hence increasing its stock of claims on the ROW. Whereas deficit means the country is reducing its net claims on the ROW. The surplus in the current account will be offset by deficit in the capital account & vice versa. The total receipts and total payments arising out of transfer of goods and services and long term capital movement.

All the transactions are regrouped under two different categories: a) autonomous transaction b) induced transaction or accommodating capital flows. Surplus in the BOP is defined as excess of autonomous receipts over autonomous payment.

The deficit is an excess of autonomous payment over autonomous receipts. That is, autonomous receipts greater than autonomous payment which results BOP surplus while autonomous receipts less than autonomous payment which results BOP deficit.

The **autonomous transactions**: that takes place independently of the BOPs. Such transactions take place for the sake of their own. i.e. on the account of people's desire to consume or to earn profit. Example, export and import of goods and services change in current account elements. Export and imports take place irrespective of other transactions entered in the BOP account. If $\text{export} = \text{import}$ in value, there will be no other transaction. If $\text{export} \neq \text{import}$, it leads to short-run capital movement, e. g, international borrowing or lending, these capital movements are not taking place for their own sake but for the sake of making payment for the deficit in the BOP.

Other autonomous elements on the current account are gifts and donations. They are voluntary and deliberate. On the capital account, the export & import of long-term capital are autonomous transactions; short-term capital movements motivated by the desire to invest abroad for higher return are also autonomous

Accommodating capital/induced/ transaction: those transactions which finance any difference between autonomous receipts or payment. These involve accommodating capital flows. Short term capital movements, gold movements and accommodating capital movements on account of autonomous transaction are induced transactions. These transactions lead to reduction in the gold and foreign exchange reserves of the country.

The total receipts and payments resulting from the autonomous transactions determine the deficit or surplus in the BOP. If receipts & payment are imbalance, BOP disequilibrium will be realized. If total payment exceeds total receipts BOP shows deficit. If a receipt from autonomous transaction exceeds payments for the autonomous transactions, BOP shows surplus. If both equals no disequilibrium

Depletion of gold & foreign exchange reserves are indicators of BOP running into deficit, which calls for government concern. If reserves are plentiful and government adopt a deliberate policy to run it down, deficit in BOP is not taken as unhealthy sign of the economy. Disequilibrium Surplus is also not a serious problem unless it causes inflationary condition. However,

accumulation of large surplus of foreign currency does often cause inflation, then it become a serious concern.

Determinants of autonomous transactions are: International demand for foreign goods & services, income –elasticity of demand for imports, the price of imports and domestic substitutes, price elasticity of demand for imported goods and people's preference for foreign goods.

£ **Kinds and causes of BOP disequilibrium**

The first and foremost cause of the disequilibrium in BOP is the change in price level. Price changes may be inflationary or deflationary. Deflation causes BOP surplus. BOP surplus does not cause serious concern from the country's surplus point of view unless it leads to wasteful expenditure and mal-allocation of resources.

Inflationary conditions create deficit in BOP. A deficit in BOP results in increased indebtedness, depletion of gold reserves, loss of employment, distortions in the domestic economy and cause serious problem in the deficit country.

1. **Fundamental Disequilibrium:** inflation causes the relative change in prices of imports and exports. Exchange rate remains the same, inflation causes increase in imports as domestic prices become relatively higher than the import prices. On the other hand inflation causes export to decrease because of decrease in foreign demand for domestic goods due to increase in domestic prices. The increase in import also depends on price elasticity of demand for imports in the home market and decrease in export depends on the price elasticity of the foreign demand for domestic goods. If price elasticities are not equal to zero, the gap b/n imports –exports widen rapidly. As a result there will be deficit in the BOP. When inflationary condition perpetuates, it produces long run disequilibrium. If the size of deficit is large & disequilibrium is obdurate (persistent) it is called as a **fundamental disequilibrium**.
2. **Cyclical disequilibrium:** if price fluctuation takes the form of business cycle, most countries face depression and inflation almost simultaneously. But since the economic size of nations differ, their imports are affected in varying degrees. Deficit and surplus varies from moderate to large. The country with higher marginal propensity to import(MPI) accumulates larger deficits during inflationary phase of the trade cycle and

moderate deficit or even surplus, during the depression. Such disequilibrium are known as **cyclical disequilibria**. However knowing MPI is difficult.

3. **Structural Disequilibrium:** this is another cause of BOP disequilibrium resulting from internal and external factors to the economy. Structural changes refers to the changes w/c results from such factors as depletion of cheap natural resources, changes in technology, change in industrial structure, and change in consumers tastes and fashion, etc. Such changes incapacitates exporting country & find it difficult to face the competition in the international market, due to either **too high cost of production or lack of foreign demand**. If the size of foreign trade is fairly large, the BOPs are adversely affected by structural disequilibrium. Structural disequilibrium may also originate from the discovery of new resources w/c may invite foreign capital in large measure. Large scale capital outflow may turn the BOP position into deficit.

£ **Other Factors for temporal disequilibria**

In addition to the above factors, the followings are also important:

1. Seasonal disturbances of crop failure, particularly in the primary products.
2. Rapid growth in population leading to large scale imports of food items
3. Ambitious development projects requiring heavy imports of technological know-how, machineries and equipment
4. Demonstration effect of advanced countries on the consumption pattern of the developing economies.

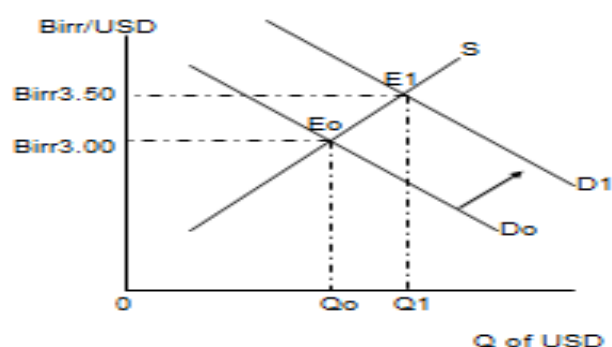
Financing BOP deficits: members with BOPs deficits may borrow money in foreign currencies, which they must repay with interest, by purchasing with their own currencies the foreign currencies held by the IMF. Each member may immediately borrow up to 25% of its quota in this way. Countries financed their deficits both by commercial borrowing and by borrowing from official agencies such as the IMF.

£ **Exchange rates and Balance of Payment**

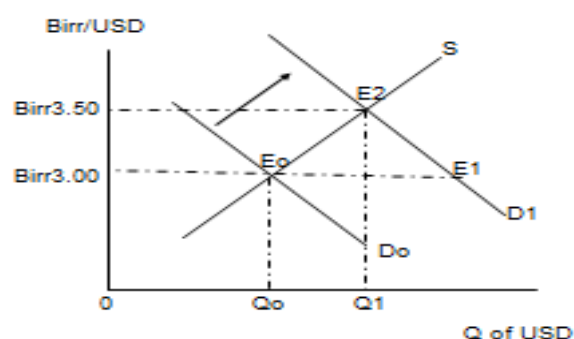
For the most part, we have been discussing a system of flexible (floating) exchange rates. Flexible exchange rate is rate determined by the forces of demand and supply without

government intervention. The rate varies according to the demand and supply of the currency in the forex market.

In the table below the floating FER is an exchange rate determined by the market forces. E_0 = Initial equilibrium. When D_d for US goods ($M \uparrow$), D_d for USD (D_0 to D_1), USD $\uparrow$ and Birr $\downarrow$. Vice Versa. This floating FER regime creates: economic instability, currency speculation, High risks for exporters and importers in the long run.

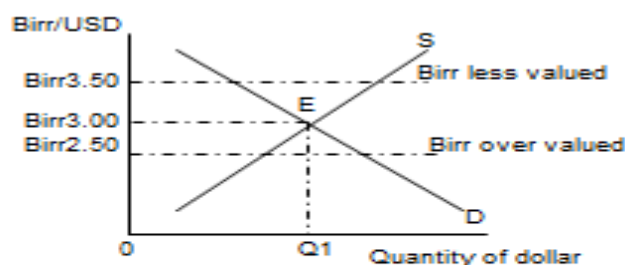


Assume that the economy is at E_0 = Initial equilibrium. When Ethiopian $\uparrow M$ from US, D_d for USD $\uparrow$ (D_0 to D_1); US BOP surplus by E_0E_1 ; USD $\uparrow$; US exports more expensive; D_d for US goods drops from E_1 to E_2 . In contrast, Birr $\downarrow$ and Ethiopia goods cheaper; US $\uparrow M$ for Ethiopia goods, $\uparrow S_s$ of USD from E_0 to E_2 . New $E_q = E_2$. Thus, Ethiopia and US BOPs are always in equilibrium between E_0 and E_2 .



Fixed exchange rate is rate of exchange between currencies pegged and maintained by the central bank's on going purchases and sales of currencies. An exchange rate determined by the gov't through buying/selling of currencies. Initial fixed exchange rate is at equilibrium at point E. If the Ethiopian Government sees this rate is not suitable, new rate can be as follows: 1) Birr less

valued which equal Birr has less market value which indicated Birr depreciation. To encourage exports and discourage imports – Birr depreciation makes Ethiopian products relatively cheaper than foreign goods. 2. Birr over valued = Birr > market value = Birr appreciation To discourage speculation on Birr (buying and selling Birr for profit).

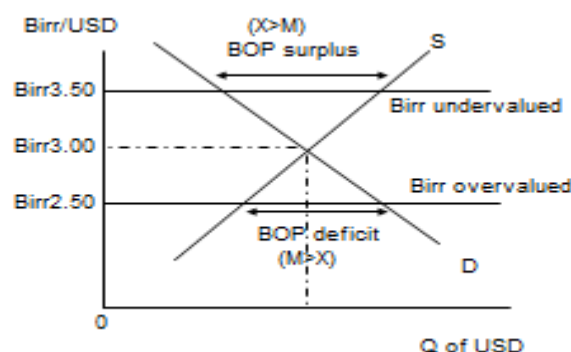


USD1=Birr3.50 or Birr1=USD0.286

USD1=Birr3.00 or Birr1=USD0.333

USD1=Birr2.50 or Birr1=USD0.400

Fixed FER would not achieve equilibrium BOP. At USD1=Birr3.50: S_s of USD > D_d for USD ($X > M$ for Ethiopia). To fix this rate, the Government uses Birr reserve to buy USD. To achieve, equilibrium, USD should ↓. In contrast, to remain at USD1=Birr2.50, the Government sell USD reserve to finance the BOP deficit (imports).



3.6. Approaches to Balance of Payments

There are three basic models that investigate the impact of exchange rate change on the current account position of the economy. These are commonly known as the elasticity approach, the absorption approach & monetary approach.

£ *Will a devaluation of the exchange rate lead to a reduction of a current account deficit?*

The answer to this question is very important because if an exchange rate change cannot be relied upon to ensure adjustment of the current account.

3.6.1 Elasticities Approach

This approach provides an analysis of what happens to the current account balance when the country devaluates its currency. The model makes some simplifying assumption; it focuses on demand condition and assumes that:- The elasticity supply of the domestic export good & foreign import good are perfectly elastic so that the changes in demand volumes have no effect on price. This assumption implies that domestic and foreign prices are fixed as a result any changes in relative price are caused by change in nominal exchange rate. This approach also assumes that capital flows occur only as a means of financing current account transactions.

The central idea of the elasticity approach is that there are two direct effects of devaluation on the current balance. One of which works to reduce a deficit, whilst the other contributes to worsen the current account deficit than before. Let us consider these two effects closely. The current account balance (CA) when expressed in terms of the domestic currency.

$$CA = PX - SP^*M \dots\dots\dots (3.1)$$

Where, P is domestic price level, X is the volume of domestic exports, S is the exchange rate, P* is foreign price level and M is the volume of import.

In other words, Current Account (CA) equals export minus import For the sake of simplicity we shall set the domestic and foreign price level at unit. The value of domestic exports (PX) will be X, while the foreign currency value of import (P*M) will be M. using these simplification equation (3.1) becomes.

$$CA = X - SM \dots\dots\dots(3.2)$$

By deviating the equation 3.1 we have

$$\partial CA = \partial X - S\partial M - M\partial S \dots\dots\dots (3.3)$$

Determining the marginal change due to small change in exchange rate is obtained by dividing 3.3.

$$\frac{\partial CA}{\partial S} = \frac{\partial X}{\partial S} - \frac{S \partial M}{\partial S} - \frac{M \partial S}{\partial S} \dots \dots \dots (3.4)$$

Now let us introduce the concept of price elasticity demand for export and import, respectively.

Price Elasticity of Demand for Export: Price elasticity of demand for export (E_x) is percentage change in export over the percentage change in price as represented by percentage change in exchange rate.

$$E_x = \frac{\partial X/X}{\partial S/S} \dots \dots \dots (3.5)$$

Price Elasticity of Demand for Import: Price elasticity of demand for import (E_m) is percentage change in import over the percentage change in price as represented by percentage change in exchange rate.

$$E_m = \frac{\partial M/M}{\partial S/S} \dots \dots \dots (3.6)$$

By substituting equation 3.5 and 3.6 in to 3.4 we have:

■ $CA = X - SM$ Deviating w.r.t the exchange rate S

$$\frac{\partial CA}{\partial S} = \frac{\partial X}{\partial S} - S \frac{\partial M}{\partial S} - M \frac{\partial S}{\partial S} \text{ divide both sides by } 1/M$$

$$\frac{\partial CA}{\partial SM} = \frac{\partial X}{\partial SM} - S \frac{\partial M}{\partial SM} - 1$$

But $E_x = \partial X/X / \partial S/S$ and $E_M = \partial M/M / \partial S/S$ thus

$$= \frac{\partial CA}{\partial SM} = E_x + E_M - 1$$

But at initial equilibrium condition, $X - SM = X/SM = 1$ then $\frac{\partial CA}{\partial S} = M(E_x + E_M - 1)$. This equation is known as the **Marshall-Lerner condition** & says that starting from equilibrium position in CA, devaluation improve the CA that is $\frac{\partial CA}{\partial S} > 0$, only if the sum of the foreign

elasticity of demand for export & the home country elasticity of demand for import is greater than unity. That is $E_X + E_M > 1$. If sum of these two elasticity is less than unity then a devaluation will lead to a deterioration of the CA.

The Role of Elasticity: The elasticities of the supply and demand for foreign exchange are fundamental determinants of adjustment to a BOPs deficit.

The Marshall-Lerner Condition: The Marshall-Lerner condition specifies the necessary condition for a positive effect of devaluation of domestic currency on the BOPs. The elasticities approach centers on changes in the prices of goods and services as the determinant of a country's BOP and the exchange value of its currency.

£ Two effects devaluation

1. **The Price Effect:** – This effect arises b/c export become cheaper measured in foreign currency. For example if coffee costs \$ 1 USD when exchange rate was 10 birr /USD, it costs only 0.5 when birr devalue to 20/1, USD. On the other hand import become more expensive after devaluation. These price effects clearly contribute to a worsening of the current account.

The **net effect** depends up on w/c effect dominates. If the increase in export volume is not sufficient to out weight a decrease in price, then less will be received from export in the foreign currency unit.

Similarly, if the decrease in import volume is not sufficient to out weight the increase in import price, then more will be paid for imports in the domestic currency. The result is that CA moves from balance to deficit.

2. **The Volume Effect:** it is much greater than the price effect. For the industrialized countries that have to face competitive export market, the price elasticity of demand for their export may be quite elastic.

The implication of the Marshall-Lerner conditions (MLC) is that devaluation may be a cure for some countries BOP deficit but not for others. There are enormous problems involved in estimating the elasticity of demand for import and export. A general consensus accepted by most

economist is that elasticity are lower in the short-run than in the long-run in which case the Marshall-Lerner condition (MLC) may only hold in the medium than the long run.

Accordingly, they considered long-run as a time period greater than two years where as short-run as a time period less than 6 months. Further, the short-run elasticity generally fails to sum to unit while the long- run elasticity almost always sum to greater than unity.

The possibility that devaluation may lead to a worsening rather than improvement in the BOP led to many researchers to find empirical evidence of the elasticity of demand for export and imports. In fact economists are divided into two groups called elasticity optimist who believed that the sum of these two elasticity tends to exceed unit. The other group called elasticity pessimists who believed that the sum of these two elasticity tends to be less than unit.

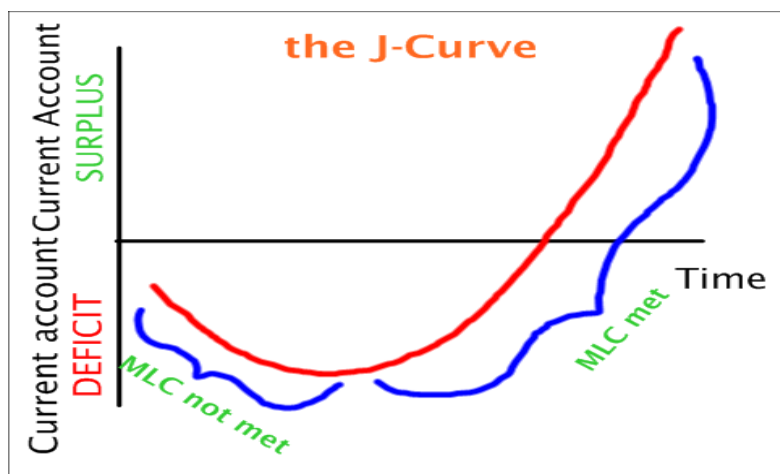
Generally, it was argued that devaluation may work better for industrialized countries than the developing countries. The justification is that many developing countries are heavily dep't upon imports so that their elasticity of demand for imports is likely to be very low. These effect of the devaluation led to the J-curve effect.

£ **Definition of 'J Curve'**

A theory stating that a country's trade deficit will worsen initially after the devaluation of its currency because higher prices on foreign imports will be greater than the reduced volume of imports. The effects of the change in the price of exports compared to imports will eventually induce an expansion of exports and a cut in imports which, in turn, will improve the BOPs.

£ **J-Curve Effect**

A devaluation of the domestic currency is unlikely to immediately improve a country's BOPs deficit. It is even possible that the devaluation could cause a country's BOP to worsen before it improves.



Why CA response is slow?

There are different explanation as to the low responsiveness, of export and import volume in the short-run and why the response is greater in the long-run. Some of the most important explanations are of the following.

1. **A time lag in the consumers response:-** It takes time for consumers in both the devaluating country and the rest of the world to respond to the changed competitive situation. Switching away from foreign imported goods to domestically produced goods takes some time because consumers will be worried about issues other than the price change such as reliability and reputation of domestic goods as compared to the foreign import, while foreign consumers may be reluctant to switch away from domestically produced goods towards the exports of the developing countries.
2. **A time lag in producer, response: -** Even if devaluation improves the competitive position of exports, it will take time for domestic producers to expand production of exportable goods. Moreover, orders for imports are made well in advance & such contracts are not readily cancelled in the short-run. Factories will be reluctant to cancel orders for vital inputs and raw materials.
3. **Imperfect Competition: -** penetrating the foreign market & building market share in the foreign market is not an easy operation & is a time –consuming & costly business. Due to this fact foreign exporters may be reluctant to loss their market share in the developing countries & might respond to the loss in their competitiveness by reducing their selling price, to the extent that they do this in the face of the rise in the cost of import caused by the

devaluation. Similarly, foreign importers may react to the threat of increased export by the devaluating country by reducing prices in their home markets, limiting the amount of additional export by the devaluating country. The existence of imperfect competition gives foreign firms some excess profit margin enabling them to reduce their price. If foreign firms were in a high competitive environment they would only be making normal price & so would be unable to reduce their price. In addition to the above effect it is unlikely that the price of exports as measured in the domestic price will remain fixed.

4. **Many imports are used as inputs for exporting industries**, and the increased price of imports may lead to higher wage costs as workers seek compensation for higher import prices; this will to some extent lead to a rise in the export price reducing the competitive advantage of devaluation.

3.6.2. Multiplier Approach

The multiplier approach is a modified & extended version of the elasticity analysis. The exchange rate is assumed fixed. The theory is suitable to analyze the adjustment process under a pegged regime. The only possibility for BP adjustment in this model is by changes in national income. Assumptions, underemployed resources; Rigidity of all prices; Absence of capital mobility; All exports are made out of current output.

National Income: $Y = C + I + G + (X - M)$

$$C = C_0 + cY, I = I_0, G = G_0, X = X_0, M = M_0 + mY$$

Thus: $Y = \frac{1}{1-c+m} (C_0 + I_0 + G_0 + (X_0 - M_0))$

An expansionary fiscal policy (a rise in G_0), an expansionary monetary policy (a rise in I_0 resulting from lower interest rates), or added exports (a rise in X_0) can increase national income.

$$\frac{dY}{dG_0} = \frac{dY}{dI_0} = \frac{dY}{dX_0} = \frac{1}{1-c+m} > 0$$

While a contractionary fiscal policy, a contractionary monetary policy or reduced exports will decrease national income. An expansionary fiscal policy or an expansionary monetary policy can worsen a country's CA (and then its BOPs).

$$\frac{dCA}{dX_0} = \frac{1-c}{1-c+m} > 0$$

In conclusion, when an economy has underemployed resources, fiscal policy, monetary policy & trade policies can be used for adjusting its **BOPs**.

- **Contractionary fiscal or monetary policy can improve the balance of payments** but at the cost of a decrease in national output.
- Added exports resulting from export-encouraging policies will improve the balance of payments & meanwhile, increase national income.

3.6.3. Absorption Approach

The elasticity approach discussed previously was on the basis of the assumption that all other things are equal. This assumption was considered as one of the weakness of the elasticity approach. In fact, changes in export & import volumes will have implication for national income & consequently income effect need to be incorporated in a more comprehensive analysis of the effects of devaluation.

The absorption approach assumes that prices remain constant & emphasizes changes in real domestic income. Hence, the absorption approach is a real-income theory of the BOP. The analysis of such evaluation was given by Alexander (1952). His analysis was focusing on the fact that a CA imbalance can be viewed as the d/c b/n domestic output & domestic spending.

Domestic spending like $(C+G+I)$ are called **Absorption**. Consider the national income identity.

$$Y=C+I+G+X-M$$

Absorption: $A = C+I+G$

National Income: $Y= C+I+G+(X-M)$

Current Account: $CA = X-M \Rightarrow CA = Y-A$. Thus, $dCA = dY - dA$

It shows whether a currency devaluation can improve the current account (then the BOPs) depends on its effect on *national income* & on *domestic absorption*. The above equation can be interpreted as CA, represents the d/c b/n domestic output & domestic absorption.

A **current account surplus** means that domestic output exceeds domestic spending whereas a current account deficit implies that domestic spending is larger than the domestic output. Since, $CA = Y - A$, the effect of devaluation on the CA will depend up on how the devaluation affects national income relative to domestic absorption. If devaluation raises domestic income, relative to domestic absorption, the current account improves. If devaluation rather increased domestic absorption relative to domestic income the current account deteriorates. Thus, understanding how devaluation affects both income & absorption is the fundamental concept of this topic.

Absorption has two parts, these are; a rise in income w/c will lead to an increase in absorption & this is determined by marginal propensity to absorb (a) and a direct effect on absorption w/c is all the other effects on absorption resulting from devaluation & this is denoted by - A.

Change in total absorption is a sum of change in income that affects absorption (a. dY) & change in direct effect dA.

Change in current account can be shown as

$$dCA = dY - dA$$

$dCA = dY - (a dY + dA)$ and this can be rearranged as

$$dCA = (1-a) dY - dA$$

This Equation tells us the fact that three factors need to be considered when analyzing the effect of devaluation. These are, Is the marginal propensity to absorb greater or less than unity?; Does devaluation raise or lower direct absorption?; Does Revaluation raise or lower direct absorption?

The condition for devaluation to improve the current account is that $(1-a)\Delta Y > \Delta A_d$ That is, any change in income not spent on absorption must exceed any change in direct absorption. To look at whether the above condition will be full filled it is worth to analyze by separating the set of economy into below full employment so that income may rise, and full employment so that income may not rise.

£ THE EFFECT OF DEVALUATION ON NATIONAL INCOME

That is clear if the economy is at less than full employment devaluation most likely either rises or lowers the national income. If the marginal propensity to absorption is less than one, then the rise in income will raise the income to absorption ratio and so improve the current account. Whereas, if income were to fall this would raise the absorption-to income ratio which would worsen the current account.

There are two important effects on income that need to be examined. These are employment effect and the term of trade effect.

Employment effect: - If the economy is at less than full employment, the MLC is fulfilled and there will be an increase in net export following a devaluation which will lead to an increase in national income and employment through foreign trade multiplier. However, if the MLC is not fulfilled then net exports would fall, implying that national income falls. Hence, it is not clear whether the employment effect will raise or lower national income.

Terms of trade effect: Terms of trade are the price of export divided by the price of import, & can be expressed algebraically as, $TT = P/(SP^*)$

Where; P –is the domestic price index, P^* - is the foreign price index, and S – is the exchange rate (domestic currency per unit of foreign currency)

Devaluation (a rise in S) tends to make imports more expensive in domestic currency terms & if this is not matched by a corresponding rise in export price the terms of trade deteriorate. Deterioration in the terms of trade represents a loss of real national income b/c more units of exports have to be given up to obtain a unit of import. Hence, the terms of trade effect lowers national income.

Overall, the effect of devaluation on the income of the devaluating country is ambiguous. Even if there is increase in net export earnings, the negative terms of trade work to reduce national income.

Even if income rises overall, it is still not clear what the implication of such a rise are for the CA, this will depend up on the value of the marginal propensity to absorb. If marginal propensity to

absorb is less than unity then an increase in income leads to an improvement in the current account b/c income rises by more than absorption. However, if the marginal propensity to absorb is greater than unity then the increased income would lead to an even bigger rise in absorption resulting in a worsening of the current account.

£ THE EFFECT OF DEVALUATION ON DIRECT ABSORPTION

Let us assume that the net effect of devaluation on the national income is zero. If devaluation reduces direct absorption then it will lead to an improvement in the current account. Whereas if direct absorption increase, then the current account deteriorate. Now let us consider possible ways in which devaluation can be expected to have impact on direct absorption.

Real balance effect:-A simple formulation of demand for money is a demand for real money balance. If price doubles then agents will demand twice as much money as before. The money demand function can be shown as, $\frac{M}{P_I} = K$, Where, K represents some constant & P_I is an aggregate price index w/c is given by,

$$P_I = \sigma P + (1 - \sigma)SP *$$

Where, σ -the percentage of expenditure on domestic goods, P – the price of domestic goods, P^* - the price of imported goods, and S- the exchange rate defined as domestic currency per unit of foreign currency.

Example, Suppose that the price of the domestic good is birr 10 while the price of the foreign good is \$2, & the pre-devaluation exchange rate is birr 10/1 USD. The domestic consumers spend 60% of their income on domestic goods so that $\sigma=0.6$. Then the average price level can be determined as;

$$\begin{aligned} P_I &= 0.6 \times 10 + (1 - 0.6) \times 10 \times 2 \\ &= 6 + (0.4 \times 10 \times 2) = 6 + 8 = 14 \text{ birr} \end{aligned}$$

If the birr is devaluated to say birr 12/1USD, then the average price level will be changed as, $P_I =$

$$\begin{aligned} &0.6 \times 10 + 0.4 \times 2 \times 12 \\ &= 6 + 9.6 = 15.6 \text{ birr} \end{aligned}$$

This means, a 20 percent devaluation of birr will raise the average price index facing Ethiopia consumers, by approximately 11.23 percent. Given an exchange, money stock & the assumption that economic agents aim to maintain a given amount of real money balance as; $\frac{M}{P_I} = K$. The

devaluation of domestic currency means that economic agents have to maintain their **real balance by cutting down on direct absorption**. Economic agents will attempt to increase their money balance by selling bonds, w/c pushes down the prices of bonds raising the domestic interest rate. The rise in interest rate will reduce investment and consumption, so **reducing direct absorption**. For the real balance effect to work the authorities must not accommodate the increase in money demand by a corresponding increase in the money supply. If the money supply increases, it will leave the ratio M/P_I constant so that the real balance effect will not be effective.

Income Redistribution Effect:-This effect arise due to the fact that **devaluation of domestic currency may increase general price index**, and this most likely have a number of effects on income distribution. To the extent that it redistributes income from those with a low marginal propensity to absorb to those with a high marginal propensity to absorb, this will increase direct absorption. There are some scenarios with this respect. These are:

- A. **Increase in the general price index** will tend to reduce the real income of those with fixed incomes. It is believed that the groups on fixed incomes are poor who have a higher propensity to absorb. Income is redistributed from those with fixed income to those with variable incomes; this income redistribution effect will tend to reduce direct absorption.
- B. **Devaluation** commonly leads to an improvement of company profits through increased sales in export & increased sale of import competing industries. The effect on direct absorption of this redistribution is not clear: while firms may have a tendency to lower absorption than workers this will be very much dep't on their expectations about the future. If their expectation is favorable then the devaluation & profit rise may stimulate investment and even raise direct absorption.
- C. There may be considerable income adjustment within groups of companies and workers. Some companies' profits will benefit from devaluation as export sales rise. However, Some firms that are dependent on imported inputs may find their cost increased and reducing their profit margin.

The overall effect on direct absorption will then depend up on the companies that gain has a higher propensity to absorb than those that lose.

Money illusion effect:-It is also argued that even if prices are increased due to devaluation of domestic currency, Consumers may exactly consume the some quantity b/c they suffer money illusion. If money illusion is the case, consumers are actually spending more on direct absorption than before. However, the money illusion effect may work in reverse way. This is b/c, consumers due to price rises, may actually decide to cut back direct absorption by more than the proportion to the price rise, so **that direct absorption falls.**

Expectation Effect:-It is possible that economic agents regarded the increase in price due to devaluation of domestic currency may spark further price rises. Such expectation may lead to an increase in direct absorptions which would worsen the BOPs. However, it can be argued that inflationary expectation may reduce investment which may rather lower direct absorption.

Laursen-Metzler Effect:- According to this effect the deterioration in the terms of trade following devaluation of domestic currency will have two effects on absorption. These are income and substitution effect.

- **Income effect:-**The deterioration in the terms of trade lower national income and thereby income related absorption .
- **Substitution effect:-**This effect arises due to the fact that devaluation makes domestic products relatively cheaper than imported goods, which results in a substitution of domestic goods for imported goods.

If the positive substitution effect out weight the negative income effect, devaluation will lead to arise in absorption and vise versa, hence, the effect of devaluation on direct absorption are a ambiguous.

3.6.4. Monetary Approach

The analysis of BOP using a monetary approach is based on the following assumptions such as there exist a stable money demand function, the aggregate supply is fixed or vertical and PPP largely work among nations, involved in international trade.

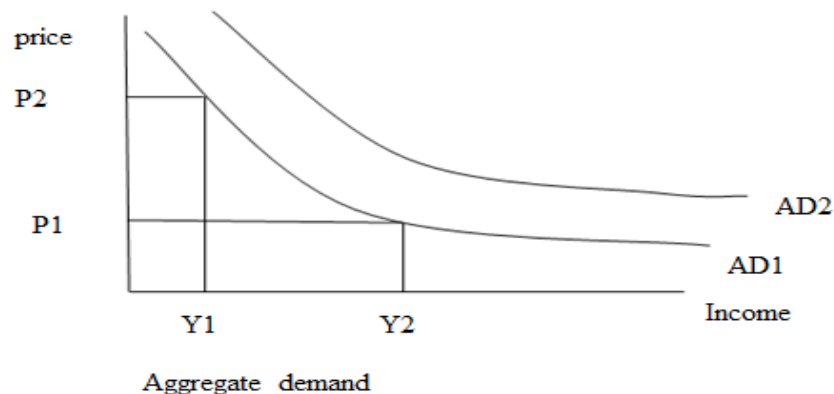
❑ Stable Money Demand Function

The monetarists use the quantity theory of money to explain stable money demand function. This is presented as:

$$Md = kpy \dots\dots 1$$

Where, Md –is the demand for nominal money balance, P -is domestic price level; Y -is real domestic income, and K -is a parameter that measures the sensitivity of money demand to change in nominal income.

The demand for money is positively related to the domestic price level, b/c as price goes up money is needed to buy the same quantity of goods & services. Moreover a rise in the domestic price level will reduce the real money balance. The real money balance is money stock over price level, i.e., M/P . The fall in real money balance will lead to an equi-proportionate increase in the demand for money. The demand for money is also positively related to real domestic income. The increase in real income will lead to increase in the transaction demand for money. On the basis of equation 1 we can depict the aggregate demand for a simple a monetary model. From equation (1) we held fixed money demand & assume that the parameter k is also fixed. We further assume that money supply is also fixed.



From this figure we can observe that an increase in real income(Y) from Y_1 to say, Y_2 require equi-proportionate fall in the price level from P_1 to P_2 . Hence $P_1Y_1 = P_2Y_2$ since the parameter K is fixed by assumption. A fall in the price level from P_1 to P_2 given a fixed money supply will increase the real money balance (M/P) & this leads to increase in aggregate demand from Y_1 to

Y2. An increase in the money supply will shift the aggregate demand to the right from AD1 to AD2.

£ Vertical Aggregate Supply curve

The simple monetary model assumes that the economy is always at full employment as the labor market is sufficiently flexible. Wage rates are flexible that they are constant at the level that equates the supply & demand for labor. Moreover, a rise in the domestic price level does not lead to an increase in domestic output b/c wages adjust immediately to the higher price level so that there is no advantage for domestic producers to take on more labor. This means that the aggregate supply is vertical at full employment level of output.



Aggregate supply: Even if the aggregate supply is fixed at full employment level Y1, it does not mean that output is always fixed. The aggregate supply may shift to the right say to AS2 if there is an improvement in productivity due to technological progress.

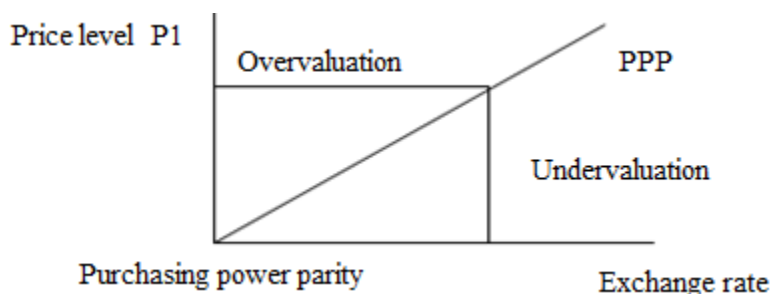
Purchasing power parity (PPP)

One of the assumptions of the monetary model is that the purchasing Power Parity (PPP) will hold among economies. The theory of PPP is explained in chapter 2. However, let's revisit some basic concepts.

According to this theory the exchange rate adjusts so as to keep the following relationship:

$$S = \frac{P}{P^*} \quad \text{that is} \quad P = SP^*$$

Thus, PPP shows the relationship b/n domestic price & foreign price level. In other words, it shows combination of the domestic price level & exchange rate w/c are compatible with PPP given the foreign price level P^* .



PPP curve has a slope given by P^* & implies that $X\%$ rise in the domestic price level requires an $X\%$ depreciation of the home currency to maintain PPP. Points to the left of PPP curve represents an **over valuation of the domestic currency in relation to PPP**, whereas points to the right represent **undervaluation in relation to PPP**.

Using these three assumptions of the monetary model & some accounting identities, we can develop a theory of BOP. The **domestic money supply** is composed of two components, namely **domestic bond holdings & the foreign currency reserve** valued in the domestic currency. This can be:

$$Ms = D + R \dots \dots 2$$

Where, M_s is the domestic money base; D is the domestic bond holding of the monetary authority; R is the foreign reserve holdings.

Accordingly, the domestic monetary base comes into circulation in either of the following ways: The monetary authority may conduct an open market operation (OMO) to purchase bonds held by private agents by the NBE. This increases NBE's liabilities but increases its asset of domestic bond holdings as represented by D .

The authority may conduct foreign exchange operation (FXO) w/c is a purchase of foreign currency asset (money or foreign bond) held by private agents by NBE. These again increase the NBE's monetary liabilities but increase its asset of foreign bonds & are represented as R .

Equation 2, state that any change in the domestic money supply (monetary base) can be caused by the change either in an OMO or by the change in FXO of the monetary authority.

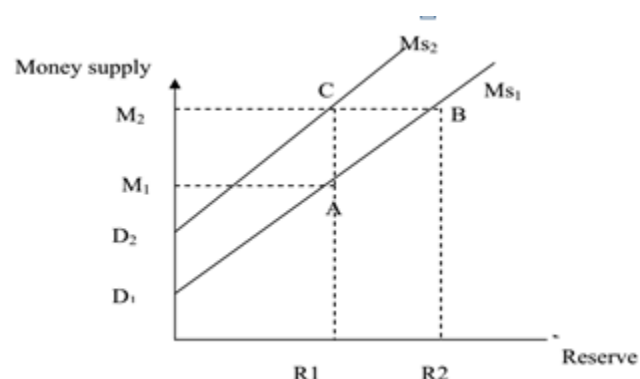


Fig.3.5. the relationship between money supply and reserve.

In the above figure when bond holdings D is fixed at D_1 all the domestic money supply is made up of entirely of the domestic component since reserves are zero at this point. For the sake of simplicity we set the exchange rate of domestic to foreign currency equals to unity. Under this condition an increase of 1 unit of foreign currency leads to an increase in the domestic money supply by 1 unit, so that when reserves are R_1 the money supply is M_1 that is, $D_1 + R_1$. The increase in domestic monetary base from D_1 to D_2 **due to an OMO** will shift the money supply from Ms_1 to Ms_2 ; as a result the total money supply rises from M_1 to M_2 .

On the other hand, expansion of the money supply **due to the purchase of foreign currency through FXO**, increases a country's foreign exchange reserve from R_1 to R_2 . This also will have an effect of increasing the money stock from point A to point B or monetary base from M_1 to M_2 .

£ A Monetary Exchange Rate Equations

Before analyzing the effect of d/t shock under fixed & floating exchange rate let's how the **exchange rate is determined in the simple monetary model**. The demand for money in the home country is given by: $Md = KPY$. The demand for money in foreign economy is given by: $Md^* = K^*P^*Y^*$; where Md^* = foreign money demand; K^* = foreign nominal income

elasticity of demand for money; P^* the foreign price level & Y^* real foreign income. The exchange rate determined by PPP & is shown by $S = \frac{P}{P^*}$.

In equilibrium money demand is equal to the money supply in each country & can be shown as: $Md = Ms$ and $Md^* = Ms^*$ The relative money supply function can be expressed as:

$$\frac{Ms}{Ms^*} = \frac{KPY}{K^*P^*Y^*}$$

From PPP; $S = \frac{P}{P^*}$ the above equation can be reduced to:

$$\frac{Ms}{Ms^*} = \frac{sKY}{K^*Y^*}$$

Solving for exchange rate we have: $S = \frac{Ms/Ms^*}{KY/K^*Y^*}$

The above last equation says that the **exchange rate is determined by the relative supply & demand** for the d/t national money stock & increase in foreign income relative to domestic income will lead to a depreciation (rise) of the domestic currency, while an increase in domestic income relative to foreign income leads to an appreciation (fall) in the exchange rate.

This is because an **increase in domestic income leads to an increased transactions demand for domestic currency → appreciation**. Based on this model of exchange rate determination, we can analyze the effect of money supply, income changes & changes in foreign price level.

£ A MONEY SUPPLY EXPANSION UNDER FIXED EXCHANGE RATES

When the exchange rate of a country's currency is fixed, the monetary authorities have to buy the currency when it is in excess supply & sell it when it is in excess demand in the private market to avoid a currency devaluation or Revaluation.

Selling of the domestic currency by monetary authorities lead to a rise in their reserves of foreign currency. If the authorities buy the domestic currency this lead to the fall of the foreign reserve.

Let us look at what will happen if there is an expansionary open market operation (OMO) under a fixed exchange rate w/c raise the money supply by a central bank purchase of domestic

Treasury bond. The effect of such an expansionary open market operation(OMO) is depicted in figure below.

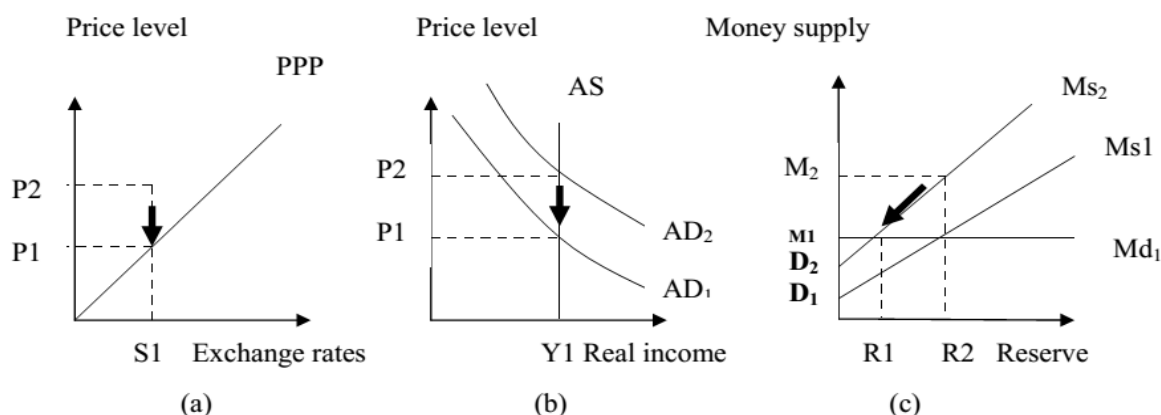


Fig.3.6. **A monetary expansion under fixed exchange rate**

An expansionary OMO Shifts the money supply curve from Ms_1 to Ms_2 & increases the domestic money supply from M_1 to M_2 . And this increase the domestic component of monetary base from D_1 to D_2 . The immediate effect is that domestic residents have excess real money balance ($M_2/P_1 > M_1/P_1$). That is, the money supply M_2 exceeds money demand M_1 . To reduce their excess real balance residents increase aggregate demand for goods w/c is given by a shift of the aggregate demand curve from AD_1 to AD_2 and this puts upward pressure on prices of domestic goods whose prices raise from P_1 to P_2 . At P_2 & fixed exchange rate S_1 , the domestic economy is not competitive in relation to PPP as it finds itself to the left of the PPP curve. The uncompetitive nature of the economy moves the BOP into deficit.

To prevent a devaluation of the currency the authorities have to intervene to purchase the domestic currency & the authorities' reserves start to decline below R_1 . The purchase back of the domestic money supply starts to reduce the excess money balance & at the same time aggregate demand starts to shift back from AD_2 to words AD_1 . As the excess money balance are reduced & this puts down word pressure on the domestic price level w/c falls back to its original level P_1 so as to arrive back at PPP.

Once the money supply returns to M_1 along the MS_2 curve the excess supply of money is eliminated and the economy is restored to equilibrium. In other words in the long-run the price level, output level, and money stock return to their initial levels.

Thus, an increase in the domestic components of the monetary base from D_1 to D_2 will lead to an equivalent fall in the reserves from R_1 to R_2 . The decrease in reserve due to purchase of the home currency leads to a return of the money stock to its original level. The monetary approach regards the BOPs deficits resulting from the expansion in the money stock to be a temporary & self-correcting phenomenon.

An expansion of the money supply causes a temporary excess supply of money and a combined current and capital account deficit, which to maintain the fixed exchange rate requires intervention in the foreign exchange market that eventually eliminates the excess supply of the currency.

Two conditions under which a BOP deficit or surplus can become more than a temporary feature. The **first condition** is when the authorities practice sterilization of their foreign exchange operations. When monetary authority intervenes to purchase their currency to prevent its being devaluated there is a reduction of the monetary base.

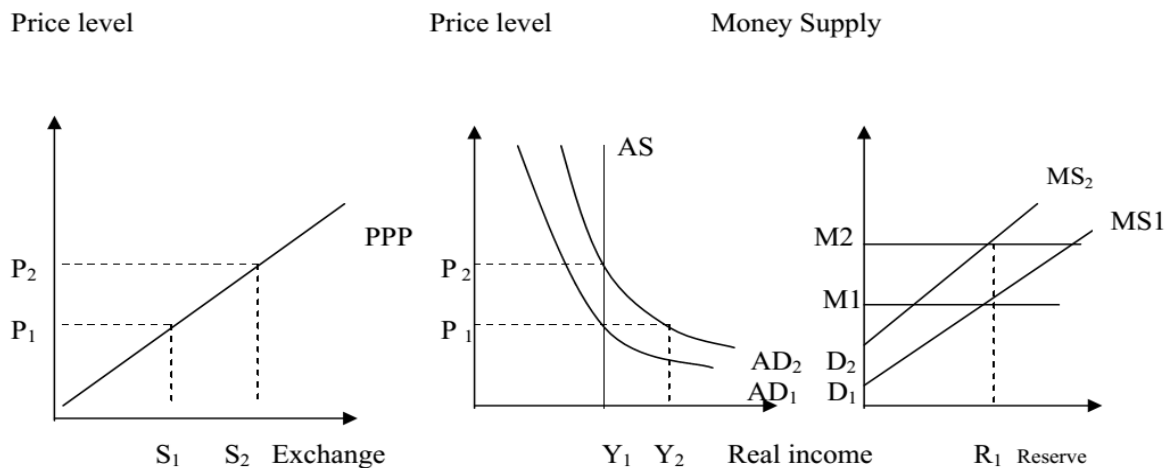
The authorities could try to offset these monetary bases by conducting a further OMO by purchase of bonds from the public. But such OMO cause a BOP deficit, which requires a further foreign exchange intervention. Hence, sterilization policy can cause a prolonged BOPs deficit, and the pursuit of such operation will be limited by the extent of a country's reserves.

Second condition that can lead to continuous deficit would be when the surplus countries were prepared to purchase the deficit country's currency & hold it in their reserve.

In such circumstances the deficit country will have its exchange rate fixed by foreign central bank intervention, & such a process can continue so long as foreign central banks are prepared to accumulate the home country's currency in their reserves. Although in this case reserve changes are zero, the deficit is reflected as an increase in liabilities to foreign authorities.

£ A Money Supply Expansion Under Floating Exchange Rates

Under floating exchange rates the monetary approach maintains that there is no as such BOP deficit or surplus & the monetary authorities do not have to intervene to purchase or sell the domestic currency in order to maintain its exchange rate. As it was shown by $\Delta BOP = \Delta CA + \Delta K - \Delta R = 0$ change in reserve is zero & any CA deficit or surplus will be offset by a net capital inflow (out flow) of the amount.



As it shown in the above figure an expansionary OMO leads to arise in the money stock from M_1 to M_2 & creates excess money balance. As a result of increase in money stock the aggregate demand shifts from AD_1 to AD_2 with demand of Y_2 which exceed domestic output. The excess demand for goods results in increase in expenditure on foreign goods & foreign investment. This leads to a depreciation of the exchange rate. As a result of the excess demand for goods the domestic price will raise leading to an increase in money demand as reflected by an upward shift of the money demand curve from Md_1 to Md_2 . When the domestic price rises demand for money will increase & this lead to a contraction of aggregate demand along the AD_2 curve & the equilibrium price P_2 is restored.

In the long-run, the effect of an $x\%$ increase in the money stock is depreciation of exchange rate by $x\%$, & increase the domestic price level by the same $x\%$. The increase in the price level induces a rise in the demand for money so that the excess money balance created by the OMO is eliminated. From equation of exchange rate, we have seen variables like M_s^* , K_y , & K^*y^* are all fixed, & any rise in M_s leads to a rise in the exchange rate S for the equation to holds true . If we compare the two exchange rate regimes (fixed and floating.) under fixed exchange rates an

expansionary OMO leads to a disequilibrium in the money market. This is solved by adjustment in the BOP & reserves held by the monetary authorities.

Under floating exchange rates an expansion in the monetary base leads to a depreciation of the exchange rate & a rise in domestic price. Moreover, the authorities can determine the amount of money supply, & money market equilibrium is restored by changes in money demand caused by change in the domestic price level & exchange rates.

In fixed exchange rates the monetary authority cannot conduct independent monetary policy like that of floating exchange rates where the authorities are free to expand & control the money supply to their desired level.

£ **CRITICISMS OF THE MONETARY APPROACH**

1. According to some economists an increase in the domestic money supply may not be reflected exclusively in the equivalent fall in the reserves under fixed exchange rates. For example, if there is unemployment (not full employment), all expansionary monetary policy may lead to some increase in output which result in increase in money demand. This may lead to slow down of the devaluation process on the home currency.
2. As to these groups of economists, the assumptions made by this approach are empirically open to question. There is ample evidence that money demand functions can be highly unstable, economies are rarely at full employment, and PPP is useless as a guide to exchange rate movements.
3. Another criticism of the monetary approach is that no attention is paid to the composition of a deficit and surplus. If there is a large deficit in the CA which is financed by an offsetting surplus in the capital account, the monetarists argue that this means there is no need for any policy concern with regard to the BOPs. Since any surplus or deficit is a transitory feature representing a stock disequilibrium in the money market which is necessarily self-correcting, a policy with regard to the BOP is unnecessary.

Review Questions

1. Define the balance of payments account and describe its significances? What are the major components of the balance of payments?
2. How are the following transactions entered into the U.S balance of payment? (a) the U.S gives \$200 cash aid to the government of Ethiopia. (b) Ethiopia uses the cash aid to import \$200 worth of machinery from the U.S.
3. Describe the approaches of balance of payment? What do elasticities of imports and exports mean? What do elastic and inelastic imports or exports imply Vis a Vis deficits and surpluses?
4. How does the absorptive approach define determination of or changes in exchange rates?
5. Which of the three approaches can be used to influence balance of payments in Ethiopia?

CHAPTER FOUR

MACROECONOMIC POLICY IN AN OPEN ECONOMY

Chapter Objectives

- Explain Open economy and macroeconomic policy
- Discuss goals of Open economy macroeconomic objectives
- Internal and external balance
- Different macroeconomics policy instruments
- International macroeconomics policy coordination

4.1. Open Economy and Macroeconomic Policy

In this chapter we will examine how both exchange-rate changes & macroeconomic policies impact upon an open economy. An economy is said to be open if it engages in unrestricted trade and commerce with other economies worldwide ,i.e. export & import.

Open economy is one that interacts freely with other economies around the world. It interacts with other countries in two ways. It buys and sells goods and services in world product markets. And also it buys and sells capital assets in world financial markets.

A fundamental difference between an open economy and a closed economy is that overtime a country has to ensure that there is balance in its current account The need for economic policy makers to pay attention to the implication of changes in monetary and fiscal policy on the BOP is an important additional dimension for consideration in the formulation of economic policy in an open economy.

4.2. Macroeconomic Policy Goals in an Open Economy

These important macroeconomic goals objectives were: Internal balance:- means (1) full employment, (2) low inflation, and (3) high economic growth. External balance: is the achievement of neither BOP deficits nor BOP surpluses.,i.e., ensuring a sustainable BOPs position over time.

£ THE PROBLEM OF INTERNAL AND EXTERNAL BALANCE

Between the years 1948 and 1973 the international monetary system was that of fixed exchange rate with the major currencies being pegged to the US dollar. Only if there are fundamental disequilibrium that authorities were allowed to devalue or revalue their currency. This implies that there was considerable interest in the relative effectiveness of fiscal & monetary policies as a means of influencing the economy.

Though economic policy makers generally have many macroeconomics objectives, the focus in the 1950, and 1960, was primarily concerned with two objectives, these are achieving full employment for the labor force and a stable level of price level.

Achieving a full employment with a stable price level is called internal balance. Even if the gov'ts were generally committed to achieve full employment, it is widely believed & recognized that expanding output in an open economy will have impact on the BOP. For instance, expanding output & employment will result in greater expenditure on imports & consequently will lead to a deterioration of the current account.

When authorities decided to maintain fixed exchange rate, then they were interested in running equilibrium in the BOP, that is, balance in the supply & demand for their currency.

The objective of having balance or equilibrium in the balance of payment is called external balance. Changes in **fiscal & monetary policies** w/c aim to influence the level of aggregate demand in the economy is called **expenditure changing policies**. Whereas policies such as **devaluation or revaluation of the exchange rate** in order to influence the composition of spending b/n domestic & foreign goods are called **expenditure switching policies**.

Many literatures, in the 1950, and 1960, were concerned with how the authorities might simultaneously achieve both internal & external balance. In fact the policy problem of achieving both internal & external balance was first conceptualized by Treuor Swan (1955) in a diagram called **Swan diagram**.

In the swan diagram below, the vertical axis represents the real exchange rate, defined as domestic currency units per unit of foreign currency so that a rise represents a real devaluation which implies improved international competitiveness. The horizontal axis is the amount of real

domestic absorption which represents the sum of consumption, investment and government expenditure

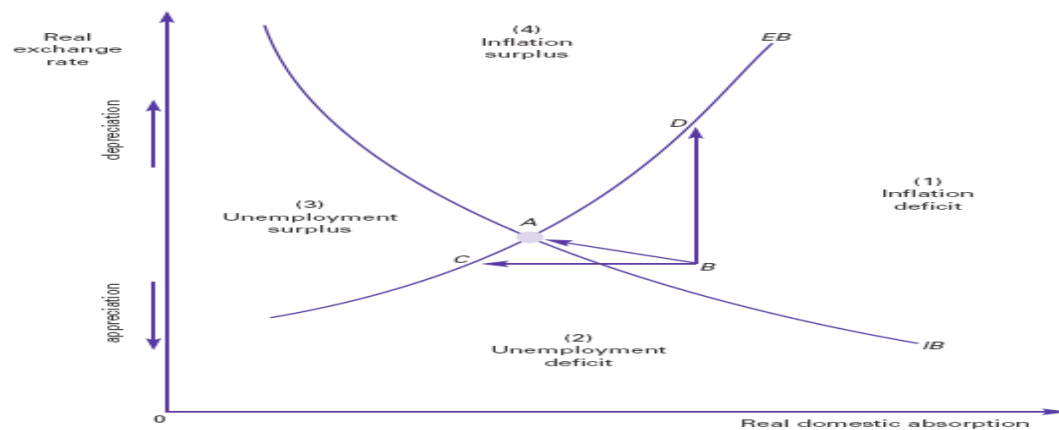


Fig.4.1. Swan Diagram

The Internal Balance (IB) schedule represents combinations of the real exchange rate and domestic absorption for which the economy is in internal balance; that is, full employment with stable prices. The IB is downward sloping. This is because an revaluation of real exchange rate will reduce export and increase imports. Thus, to maintain full employment it is necessary to increase domestic expenditure

The external Balance (EB) schedule shows combinations of the real exchange rate and domestic absorption for which the economy is in external balance that is, equilibrium in the current account. The EB schedule is upward sloping from left to right. This is because a devaluation of the exchange rate will increase export and reduces imports. So to prevent the CA moving into surplus requires increased in domestic expenditure to induce an offsetting increase in imports.

To the right of the EB schedule domestic expenditure is greater than that required for CA equilibrium so that the result is a current account deficit, while to the left there is current account surplus. The swan diagram is divided into four zones representing different possible states for the economy. Zone 1- Represents a deficit and inflationary pressure, Zone 2- Represents a deficit and deflationary pressure, Zone 3- Represents a surplus and deflationary pressure, and Zone 4- Represents a surplus and inflationary pressure.

The economy is in both internal & external balance only at point A where IB and EB schedule intersects. Suppose that the economy is initially at point B in Zone 1, experiencing both inflationary pressure and a current account deficit. If the authorities maintain a fixed exchange rate & try to reduce the CA deficit by cutting back real domestic expenditure they move the economy towards point C. Alternatively, authorities may try to tackle the deficit by devaluing the exchange rate, this has the effect of moving the economy towards point D on the EB schedule.

The important lesson obtained from this simple model is that, the use of one instrument, be it fiscal policy or devaluation to achieve two goals- internal and external balance, may not be successful. To move from point B to point A, the authorities need to deflate the economy and undertake devaluation by appropriate amount. The deflation combined with devaluation will control inflation and improves the current account so that the two objectives can be met. The idea that a country generally requires as many instruments as it has targets was elaborated by Noble prize winning Dutch economist known as Jan Tinbergen (1952)., and is known as **Tinbergen's Instruments –targets rule.**

£ Criticisms of Swan Diagram

The Swan diagram provides a useful conceptual frame-work for economic policy discussion but:

- 1) it is simplistic so that the underlying economic relationships are not explicitly defined.
- 2) There is no role for international capital movements that were an increasingly important feature of the post-world war II international economy.
- 3) No distinction is made b/n monetary & fiscal policies as means of influencing aggregate demand & output in the economy.

The Mundell -Fleming model, however, will integrate such feature into a formal open economy macroeconomics model.

£ Mundell- Fleming Model

This Model has its own origin to paper published by James Fleming (1962) and Robert Mundel (1962-1963). Their major contribution was incorporating international capital movements into formal macroeconomic models based on the Keynesian IS-LM frame work.

Derivation of IS schedule for an open economy

IS schedule for an open economy shows various combinations of output & interest rate that result in equilibrium in goods market or that makes leakages equal to injections. Where leakage=S+M, Injections=I+G+X

Goods market is at equilibrium when quantity of goods & services demanded are equal to quantity supplied or injections in the system equals to leakages in the system. In open economy we have the following identity called open economy identity

$$Y = C + I + G + X - M \quad \dots\dots\dots 1$$

We know that $S=Y-C$ therefore equation 1 can be restated in terms of leakages and injections

$$S + M = I + G + X \quad \dots\dots\dots 2$$

For simplicity the following linear r/ships are assumed

$$S = S_a + sY \quad \dots\dots\dots 3$$

$$M = M_a + mY \quad \dots\dots\dots 4$$

Investment is assumed to be Inverse function of interest. Thus,

$$I = I(r) \quad \quad \quad dI/dr < 0$$

In the diagram below, Quadrants I presents the relationship between leakages and nations income. It is upward because increase in income leads to increase in saving and imports. The resulting volume of leakages is transferred to quadrant II which has 45 degree line that converts any distance along the vertical to an equivalent distance into equivalent of injections(In),

Injection schedule is depicted in quadrant (III) which shows the given price level and state of expectations the rate of interest that leads to level of injections In1,if the interest rate is r1. This means that at Y1 leakages is L1 & Injection is In1, and this occurs at interest rate r1.

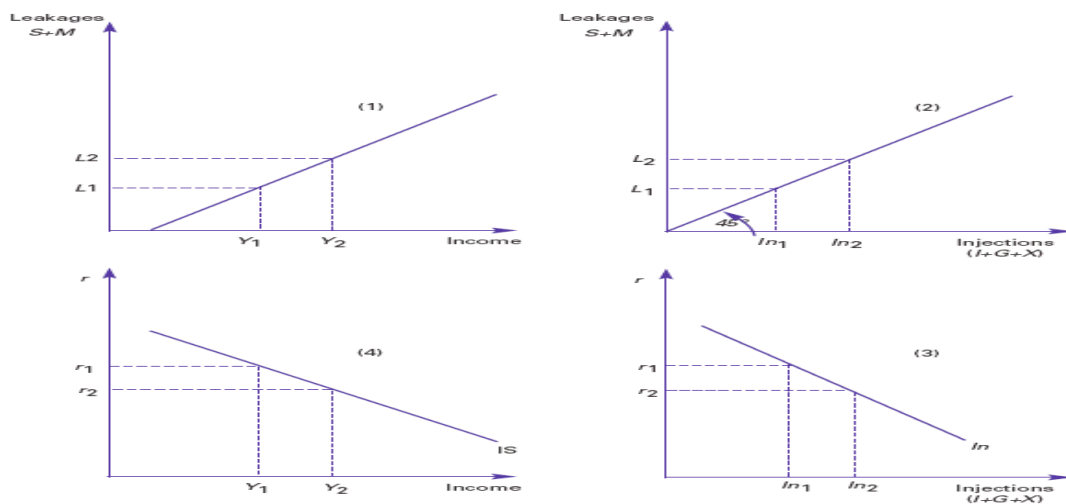


Fig.4.2. **Derivation of IS model**

In quadrant(4), we can depict a point on IS curve for an open economy because interest rate r_1 and income level Y_1 , we know that leakages are equal to injections. We repeat the same process for income level Y_2 to obtain the rate of interest rate r_2 for which leakages are equal to injections.

Derivation of LM Schedule

LM-curve show various combination of the level of income & interest rate for w/c money market is at equilibrium ($M_d = M_s$). It is assumed money is demanded for

I . Transaction Purpose—+ve function of income

$$M_t = M_t(Y)$$

II. Speculative purpose— -Ve function of interest rate

$$M_{sp} = M_{sp}(r)$$

Equilibrium in money market is : money demand equals money supply

$$M_d = M_{sp} + M_t = M_s$$

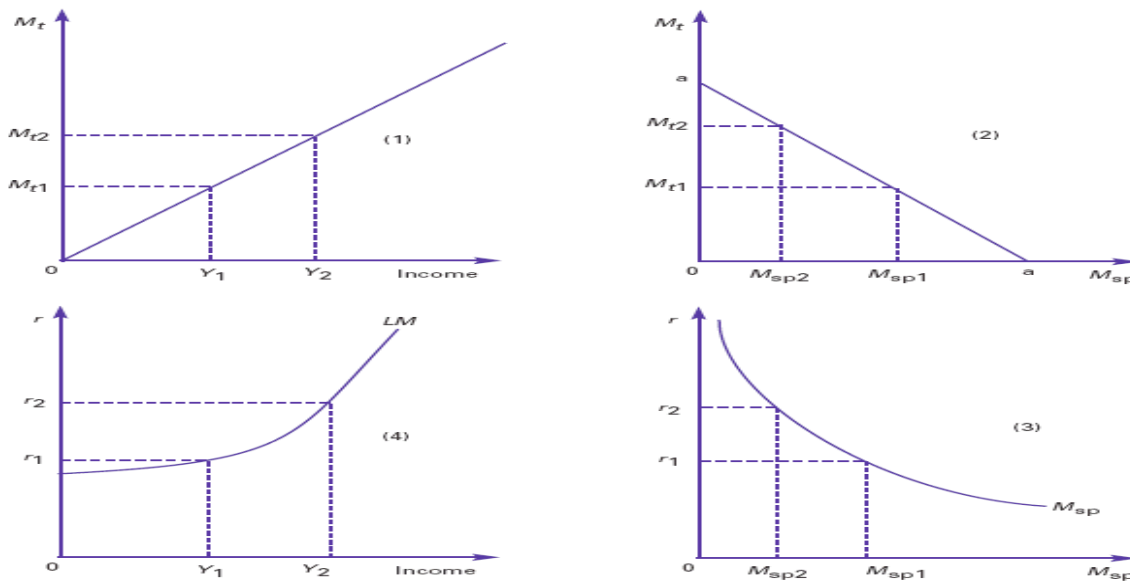


Fig. 4.3. Derivation of LM Schedule

Quadrant 1 shows the positive relationship b/n income and demand for money for transaction purpose. Quadrant 2 shows the distribution of the fixed money supply between transaction and speculative balance. Quadrant 3, show the negative relationship between interest rate and speculative demand. Quadrant 4 shows interest rate and income combination that can maintain the equilibrium of the money market.

The LM schedule is up ward sloping from left to right because high income levels imply relatively larger transaction demand, which for a given money supply can only be drawn out of speculative demand by relatively high interest rate. This is so, most of the time transaction demand for money outweigh the speculative demand for money. Regardless the level of interest rate.

£ Derivation of BP Schedule

BP schedule shows different combinations of interest rate and income that keep BOP at equilibrium.

1. Current Account: X- $M = M_a + mY$

2. Capital Account: $K = K(r - r^*)$

Net capital flow (K) is a positive function of domestic interest rate. Equilibrium in BP is: $X - M + K = 0$. This assumes position of reserve is regulated by market forces, thus the $\Delta R = 0$. This implies that if there is a current account deficit there should be an offsetting surplus in capital account to have equilibrium in BP and vice versa.

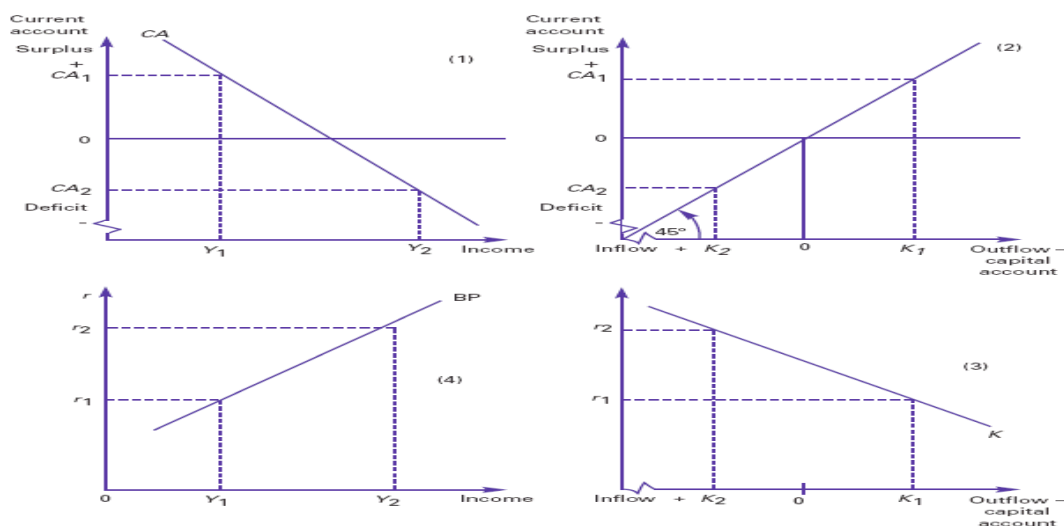


Fig.4.4. Derivation of Balance of payment

Quadrant 1 shows the negative relationship between CA and national income. Which implies that higher income causes the CA to deteriorate? Quadrant 2 transfers the CA position into an equal capital flow of opposite sign. With CA surplus CA₁ there is a required capital outflow K₁ to ensure BP equilibrium. Whereas deficit CA₂, requires a capital inflow of K₂. Quadrant 3 the interest rate that is required for a given capital flow. The capital flow schedule is down ward sloping b/c high domestic interest rate encourages net capital inflow and low interest rate encourage net capital out flow. Quadrant 4 the BP schedule is upward sloping. This is because higher level of income causes deterioration in the CA and this necessitates reduced capital outflow or higher capital inflow which requires higher interest rate. Thus ever point of the BP shows a combinations of domestic income and interest rate for w/c the overall BP is in equilibrium.

At points to the left of the BP schedule BP is in surplus because for a given amount of capital flows the CA is better than that required for equilibrium as the level of income is lower. To the

right of BP the BP shows deficit as income is higher than that compatible with overall equilibrium.

Note that the slope of BP is determined by the relative degree of international capital mobility.

If international capital mobility is very high BP will be flatter or the slope will be smaller. This is because increase in income leads to a deterioration of CA, higher degree of capital mobility and interest rate required to rise by smaller amount to attract sufficient capital inflow to ensure overall equilibrium.

When capital is perfectly mobile, a small rise in the domestic interest rate above the world interest rate leads to a massive capital inflow making the BP schedule horizontal at the world interest rate. If capital is perfectly immobile internationally then rise in the domestic interest rate will fail to attract capital inflows making the BP schedule vertical. Therefore between these two extremes, we have an upward sloping BP schedule where capital mobility is imperfect.

Equilibrium of the Model-General equilibrium of an open economy: Equilibrium occurs when IS, LM and BP schedules intersect each other at common point.

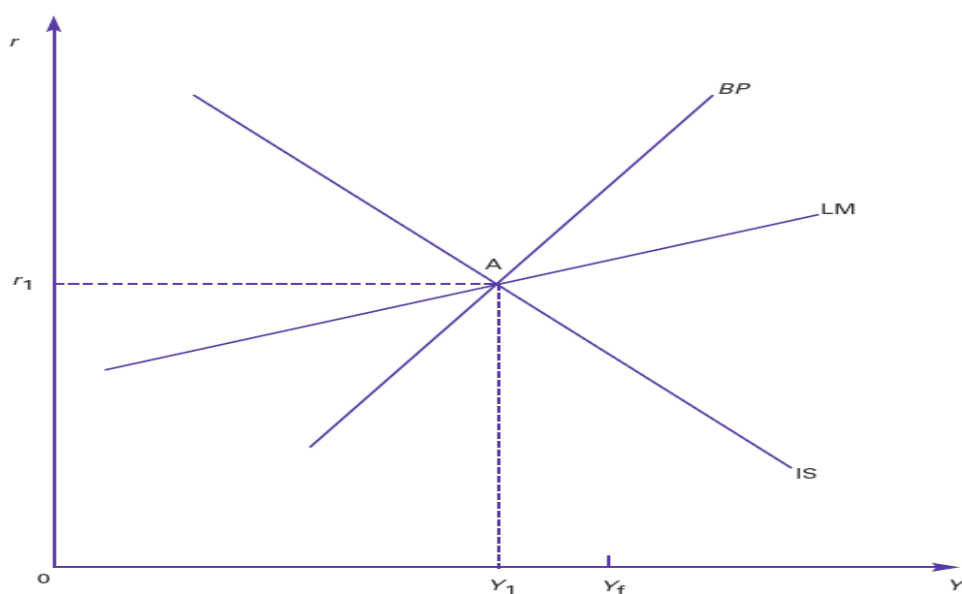


Fig.4.5. General equilibrium of an open economy

Equilibrium is achieved at interest rate r_1 and income Y_1 , the income is less than full employment (Y_f), implying that there is some unemployment in the economy.

The reason why IS-LM schedules do not intersect at the full level of income(Y_f), is that Y_f is that planned leakages ($S+m$) would exceed planned injections($I+G+X$). This would imply a buildup of stock of unsold goods leading producers to reduce output. Only at output level Y_1 do planned leakages equal to planned injections so that changes in stocks are avoided.

£ Factors shifting the IS-LM-BP schedules

Change in fiscal and monetary policies will affect the equilibrium, as change in these policies influence both income level & interest rate. Expansion in investment, government spending and export shift the IS schedule. Resulting in higher interest rate which is not compatible with the equilibrium in BP and LM. Autonomous fall in leakages (saving + imports) shift IS outwards.

Shifting Factors for LM Schedule: Increase or decrease in domestic money supply, which will have opposite effect on equilibrium interest rate and income. Depreciation of exchange rate will lead to rise in domestic price index. This implies rise in the price of imported goods this in turn reduces real money balance that can only be eliminated by reducing the transaction demand for money. This implies a lower income and leftward shift of the LM schedule.

Shifting factors for BP: Autonomous increase in export or decrease import. This improves the CA, requires right ward shift in BP. This rises domestic income at each level of interest rate. Devaluation/revaluation, devaluation if MLC is satisfied will improve the CA hence shift the BP to the right. As it increases national income Since revaluation mostly cause deterioration in the current account it shifts the BP to the left. Reduces national income

4.3. Stabilization Policy Instruments

Swan diagram revealed that use of both expenditure-switching and Expenditure changing policies by the authorities can lead to achievement of internal and external balance. But they did not distinguish fiscal and monetary policies which are quite independent and different type of expenditure changing policies. But this model distinguishes fiscal & monetary policy.

The Question is whether or not it is possible /feasible to achieve the twin objectives (internal and external balance) by combining fiscal and monetary policy without the need to adjust the exchange rate?

I. Monetary Policy

Expansionary Monetary Policy: When the authorities conduct an this policy, they purchase bonds from the public, this pushes up the price of bonds and leads to a fall in the domestic interest rate, which in turn stimulates investment and leads to a rise in output. As far as the BOPs is concerned, the increase in income leads to a deterioration of the CA and lower interest rate will lead to increase in capital outflow so that the BOP moves into deficit.

Contactionary Monetary Policy: Conversely, this policy involves the authorities selling bonds, this pushes down the price of bonds and leads to a rise in the domestic interest rate, w/c leads to less investment and a fall in output. The BOP position will improve as imports fall & the higher interest rate attracts capital inflow.

II. Fiscal Policy

Expansionary Fiscal Policy: With an expansionary fiscal policy the government increases its expenditure. Government with pure fiscal policy finances its expenditure by selling bonds. The increase in government expenditure will shift the IS curve to the right. However, the bond sales will depress the price of bonds; there by rising domestic interest rate which will offset expansion in output. The price effect of the fiscal expansion on the BOP is indeterminate b/c while the expansion of output will worsen the CA , the rise in interest rate will improve the capital account. The fiscal policy will not affect the money supply /demand in the hands of the private sector. The money raised from bond sales is re-injected into the economy, through increased gov't expenditure. **Contractionary Fiscal Policy:** converse is true

£ Sterilized intervention & non sterilized intervention

Another issue we need to distinguish is sterilized & non sterilized intervention in foreign exchange market.

- a. **Sterilized intervention:** with such intervention authorities will offset the money base through exchange market intervention to ensure that reserve changes due to intervention do not affect domestic money base.
- b. **Non sterilized intervention:** In such intervention authorities allow the reserve changes to affect monetary base.

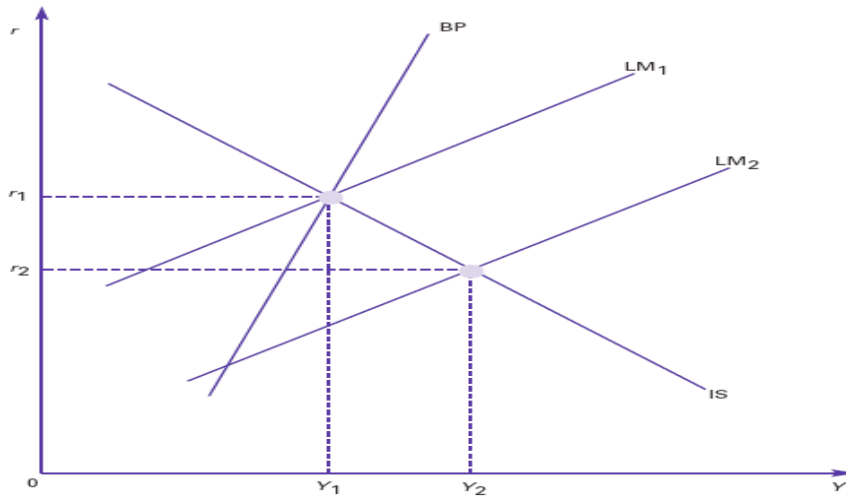


Fig. 4.7. Change in liquidity of money (LM)

Example: **Non Sterilization:** Purchasing of bonds shifts LM curve from LM1 to LM2. To maintain fixed exchange rate, authorities will purchase domestic currency by foreign currency. This reduces reserves and domestic money supply. This reduction in domestic currency is exactly offset by further expansion of money supply so that LM remains at LM2. **Sterilization:** Purchasing of bonds shifts LM curve from LM1 to LM2 to maintain exchange rate fixed, authorities will purchase domestic currency by foreign currency. This shifts LM curve back to original LM1.

4.3.1 Internal and External Balance under fixed exchange rate

Assume economy is at A with interest rate r_1 and income level Y_1 which is below full employment (level of income Y_f). Government attempts to eradicate unemployment through fiscal expansion (by selling bonds). This shifts IS curve from IS1 to IS2 which then increases output from y_1 to y_2 and internal equilibrium can be achieved at the expense of BOP balance. So authorities forced to purchase home currency in foreign exchange market by purchasing foreign exchange which shifts LM curve from LM1 to LM2 (contractionary monetary policy)

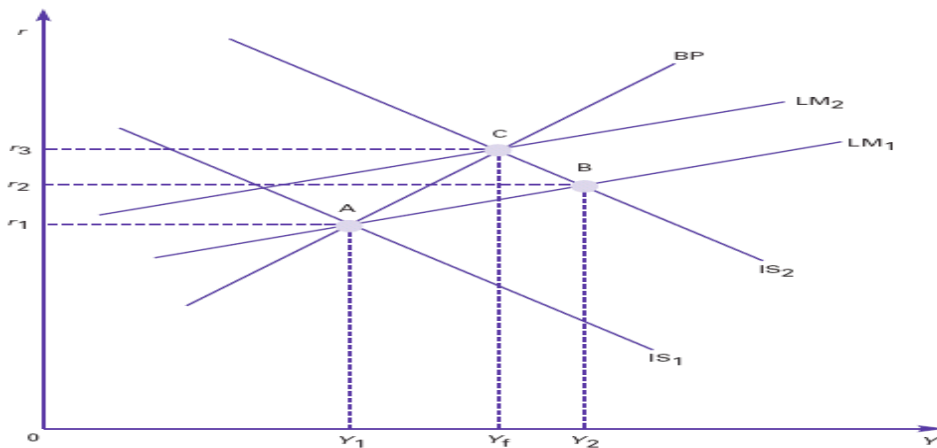


Fig. 4.9. Internal and external balance under a fixed exchange rate

Conclusion: So we can conclude that by combining expansionary fiscal and contractionary monetary policy internal & external balance can be achieved simultaneously.

4.3.2. Internal and External Balance Under Floating Exchange Rate

According to our analysis of the Swan diagram, by combining an exchange rate change with expenditure changing policy, it is possible to achieve both internal and external balance. Let us see separately fiscal and monetary expansion under floating exchange rates.

a. Monetary expansion under floating exchange rates

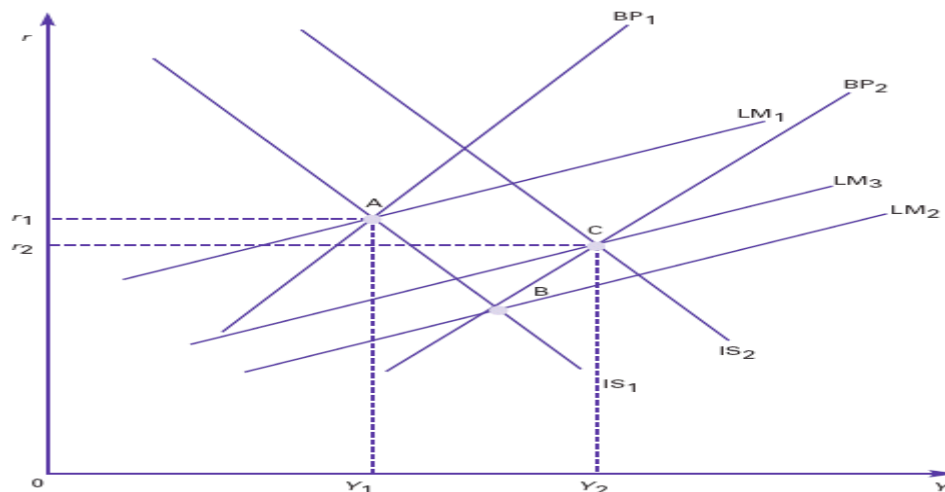


Fig.4.9. monetary expansion under floating exchange rates

Assume economy is at Y_1 (less than full employment) with interest rate r_1 . Let's assume now government adopts expansionary monetary policy and shifts LM curve from LM_1 to LM_2 w/c increases income and reduce interest rate and then deficit at point B. Exchange rate is allowed to depreciate and then this shifts IS to IS_2 and BP from BP_1 to BP_2 . However it also leads to leftward shifts of LM curve LM_2 to LM_3 until all three schedules intersect each other at common point C with new income level Y_2 and interest rate r_2 .

Conclusion: Under floating regime it is possible to achieve increase in real output to the full employment level and achieve external balance by using expansionary monetary policy with changes in exchange rate.

b. Fiscal Expansion Under floating Exchange Rate

The effect of fiscal expansion on the exchange rate under floating exchange rate depends on slope of BP schedule relative to LM schedule. Steeper BP schedule means capital flows are relatively insensitive to interest-rate changes, while money demand is fairly elastic with respect to the interest rate and reverse is true for the flatter BP schedule. So we shall consider two cases, in the first case the balance of payment- BP schedule is steeper than the LM schedule, while in case two the reverse is true.

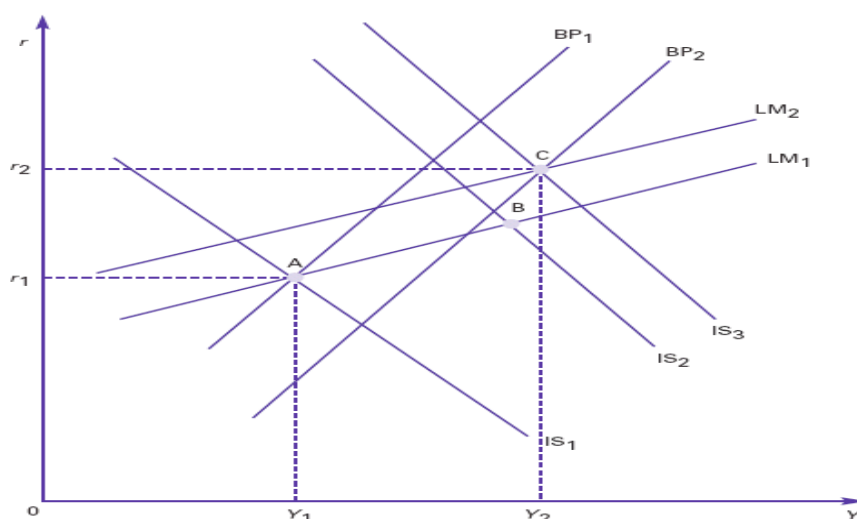


Fig. 4.10. Case 1: A fiscal expansion under floating exchange rates

Case I: Where BP schedule is steeper than the LM Schedule. An expansionary fiscal policy shifts the IS schedule from IS_1 to IS_2 which induce the rise of the domestic interest rate and domestic income. Increase in interest rate & income has opposing effects on the BOPs. That is the expansion of real output leads to a deterioration of the CA, the increase in interest rate improves the Capital account. However, since capital flows are relatively less mobile the former effect outweighs the later, as a result the BOP moves into deficit.

The deficit in the BP leads to a depreciation of the exchange rate and this has the effect of shifting the BP schedule to the right from BP_1 to BP_2 and the LM schedule to the left from LM_1 to LM_2 and the IS schedule even further to the right from IS_2 to IS_3 . Final Equilibrium is obtained at point C with interest rate r_2 and income level Y_2 . Hence, the deterioration in the balance of payment resulting from the rise in real income is offset by a combination of a higher interest rate and exchange rate depreciation.

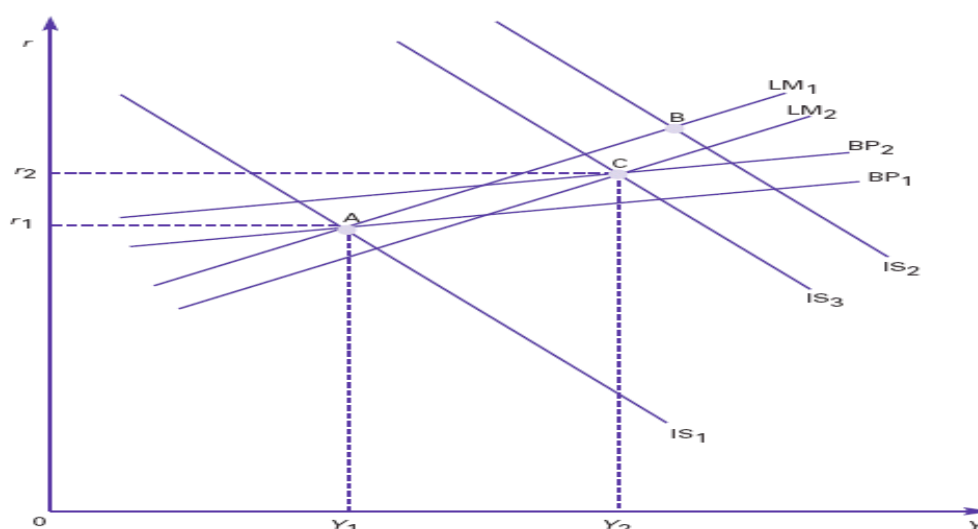


Fig. 4.11. Case 2: A fiscal expansion under floating exchange rates

An expansionary fiscal policy shifts the IS schedule from IS_1 to IS_2 . Under this case, capital flows are more responsive to change in interest rate & the BP schedule is less steeper than the LM schedule. The shift in IS schedule will raise both interest rate & domestic income. The increase in interest rate will increase the capital inflow which is more than offsets the deterioration in the CA due to the increase in income and the BOPs moves into surplus. This surplus induce an appreciation of the exchange rate which moves the LM schedule to the right

from LM1 to LM2 and the IS curve to the left from IS2 to IS3. Equilibrium is obtained at higher level of output, higher interest rate and an exchange rate appreciation.

4.3.3. A SMALL OPEN ECONOMY WITH PERFECT CAPITAL MOBILITY

One of the legacies of the post WWII is the increase in integration of international monetary system and international capital market. The desirability of capital flows and how they might endanger the ability of authorities to conduct effective economic policy has been widely discussed.

Mundell and Fleming (1962) sought to examine the implication of high capital mobility for a small country that had no ability to influence world interest rates. Their paper showed that for such a country, the choice of exchange-rate regime would have radical implications concerning the effectiveness of monetary and fiscal policy in influencing the level of economic activity.

A. FIXED EXCHANGE RATES AND PERFECT CAPITAL MOBILITY

The model assumes a small country facing perfect capital mobility. Any attempt to raise the domestic interest rate leads to a massive capital inflow to purchase domestic bonds pushing up the price of bonds until the interest rate returns to the world interest rate. Similarly, any attempt to lower the domestic interest rate leads to a massive capital outflow as international investors seek higher world interest rate. Such massive bond sale means that bond prices fall & the domestic interest rate immediately returns to the world interest rate in order to stop the capital outflow. The implication of perfect capital mobility is that the BP schedule for a small open economy becomes a horizontal straight line at the domestic interest rate which is equal to the world interest rate.

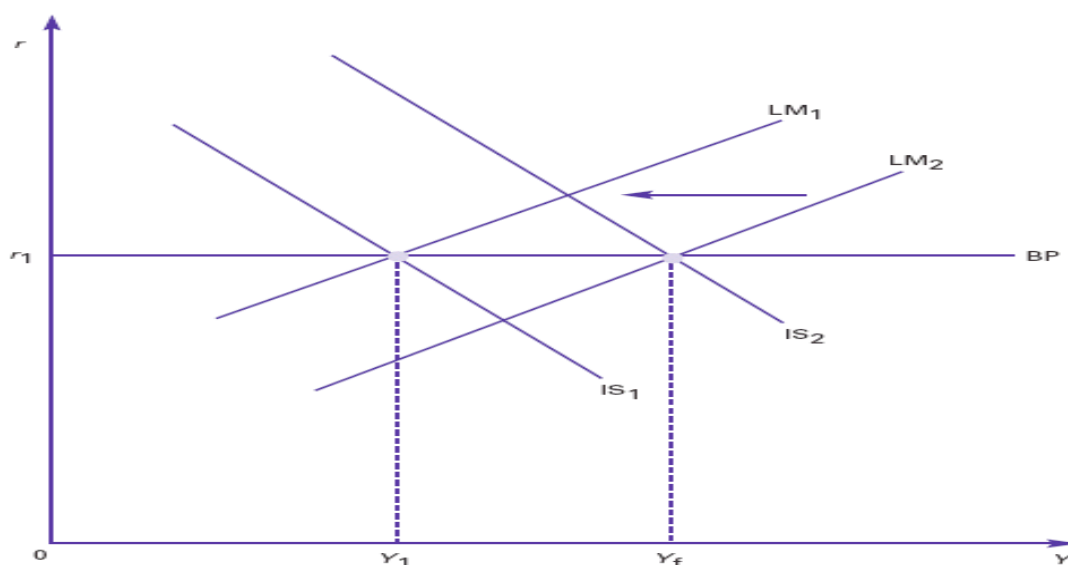


Fig. 4.12. a. fixed exchange rates and perfect capital mobility

In the figure, the initial level of income is Y_1 where IS-LM curves intersect, which is below the full employment level of income Y_f . If the authorities attempt to raise output by a monetary expansion, the LM curve shifts from LM_1 to LM_2 , and there is a downward pressure on the domestic interest rate and this results in a massive capital outflow.

The capital outflow means, there is pressure for devaluation of the currency. To prevent the devaluation authorities must intervene in the foreign exchange market to purchase the home currency with reserves. Such purchase results in a reduction of the money supply in the hands of private agents, and the purchase has to continue until the LM curve shifts back to its original position at LM_1 where the domestic interest rate is restored to the world interest rate. With perfect capital mobility, any attempt to pursue a sterilization policy leads to such large reserve loss that it cannot be pursued. Therefore, with perfect capital mobility and fixed exchange rates, monetary policy is ineffective in influencing output.

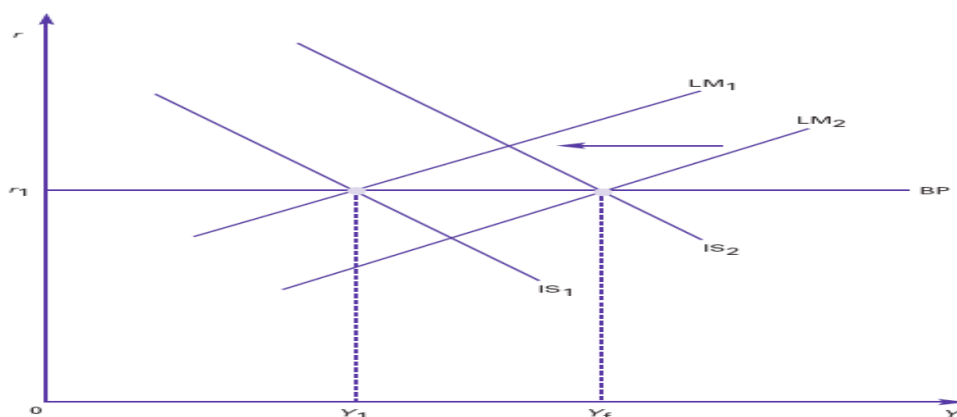


Fig. 4.12. if there is a fiscal expansion, the IS schedule will shift to the right from IS₁ to IS₂.

This shift puts up-ward pressure on the domestic interest rate w/c leads to a capital inflow.(increase in supply of FOE). To prevent appreciation of domestic currency the authorities have to purchase the foreign currency with domestic currency, this mean that the amount of domestic currency held by private agents increase. The increase in the money stock continues until the LM schedule passes through the IS₂ schedule at the initial interest rate. Thus, under fixed exchange rates and perfect capital mobility an active fiscal policy has the ability to achieve both internal and external balance. This is an exception to the instruments-target rule, although monetary policy does have to passively adjust to maintain the fixed exchange rate.

B. FLOATING EXCHANGE RATES AND PERFECT CAPITAL MOBILITY.

Suppose the economy is initially in equilibrium at income level Y₁ where the IS₁ schedules intersect the LM₁ schedule. A monetary expansion shifts the LM schedule from LM₁ to LM₂ leading to downward pressure on the interest rate, w/c in turn leads to capital outflow and a depreciation of the exchange rate.

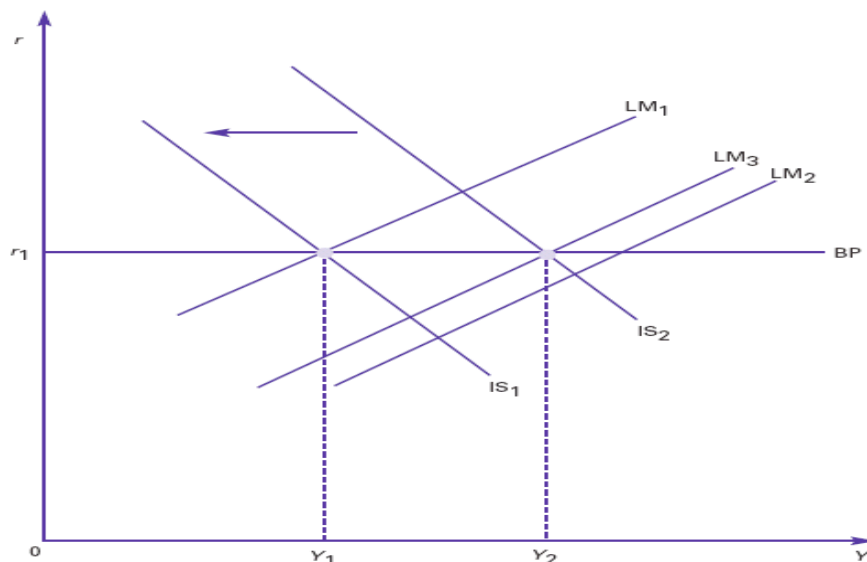
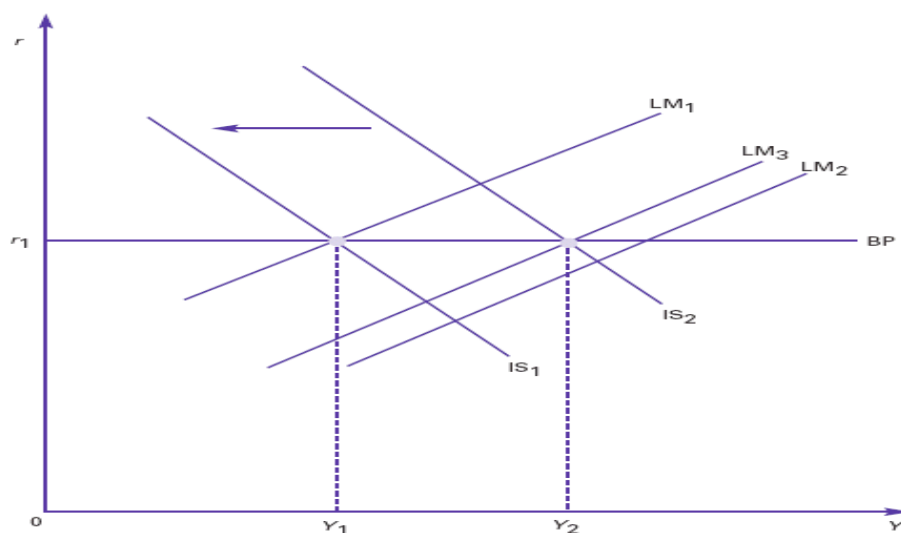


Fig.4.13. floating exchange rates and perfect capital mobility.

The depreciation leads to an increase in exports and reduction in imports. This will shift the IS curve to the right and the LM curve to the left so that final equilibrium is obtained at higher level of income say Y_2 . This implies authority could obtain both internal & external balance by monetary policy alone.



Suppose the authorities attempt to expand output by an expansionary fiscal policy. The increased government expenditure shifts the IS schedule from IS_1 to IS_2 . The increase in the sale of government bond (to finance the fiscal expansion) leads to increase in the interest rate

which leads to a massive capital inflow & an appreciation of the exchange rate. The appreciation of the exchange rate result in a reduction of export and increase in import, and this forces the IS schedule to shift back to its original position. Thus, with perfect capital mobility and a floating exchange rate, fiscal policy is ineffective in influencing output.

In summary, fiscal policy is very effective in influencing output under fixed exchange rate and monetary policy is very effective under floating exchange rate with perfect capital mobility. This is relevant to economic policy design. Under fixed rates, policy makers will pay more attention to fiscal policy than under floating rates when more emphasis will be placed on monetary policy.

4.4. International Macroeconomic Policy Coordination

What is meant by International Macroeconomic (economic-policy) Policy Coordination?

International economic-policy coordination:-is the attempt to significantly modify national policies; monetary policy, fiscal policy, exchange-rate policy in recognition of international economic interdependence. Policy coordination does not necessarily imply that nations give precedence to international over domestic concerns. Since the policies of one nation can spill over to influence the objectifies of others; nations should therefore communicate with one another & attempt to coordinate their policies.

There are many examples of international economic-policy coordination. The Smithsonian Agreement of 1971 was a coordination attempt by the major industrial nations to realign the exchange values of their currencies using currency devaluations & revaluations. The 1978 Bonn Summit resulted in the enactment by Germany & Japan of expansionary fiscal & monetary policies to stimulate their demand for U.S. goods & reduce the U.S. trade deficit; in return, the U.S raised its price of oil to the world level.

Why does the need for international Macroeconomic arise?

If economic policies in each of two nations affect the other, then the case for policy coordination would appear to be obvious. Policy coordination is considered important in the modern world because economic disruptions are transmitted from one nation to another. Without policy

coordination, national economic policies can destabilize other economies. Example: suppose the world consists of just two nations, Germany & Japan. Although these nations freely trade goods with each other, they desire to pursue their own domestic economic priorities. Germany wants to avoid trade deficits with Japan, while achieving full employment for its economy; Japan desires full employment for its economy while avoiding trade deficits with Germany.

Assume that both nations achieve balanced trade with each other, but each nation's economy operates below full employment. Both nations contemplate enacting expansionary government spending policies that would stimulate demand, output, and employment.

Potential Benefits of Policy Coordination: Policy coordination permits each nation to achieve full employment and balance of trade. In the real world however, policy coordination generally involves many countries & many diverse objectives, such as low inflation, high employment, economic growth, & trade balance.

Potential Impediments to Policy Coordination: Several obstacles hinder successful policy coordination. Even if national economic objectives are harmonious, there is no guarantee that governments can design and implement coordinated policies.

Policy makers in the real world do not always have sufficient information to understand the nature of the economic problem or how their policies will affect economies.

Policy coordination is also complicated by different national starting points: different economic objectives, different national institutions, different national political climates, and different phase in the business cycle.

Review Questions

1. Describe the internal and external balances that policy makers seek to maintain? Why need internal and external balances?
2. What should governments do when an undesirable change occurs that distort the equilibrium for an open economy? Do all economies adopt the same policy measures?
3. Suppose that an open economy with fixed exchange rate system is in equilibrium. A sudden fall in the tastes of consumers for domestic products falls. What type of policy is effective in restoring equilibrium?
4. What are some of the most important policy instruments or tools that a nation can employ to achieve important economic objectives or targets of nations? How many policy instruments or tools does a nation need to reach its objectives?
5. Suppose again that an open economy with managed floating exchange rate policy is in equilibrium. Consumptive demand for output abruptly rises. What type of policy can policy makers use to attain equilibrium?

CHAPTER FIVE

INTERNATIONAL MONETARY SYSTEM AND KEY INTERNATIONAL FINANCIAL INSTITUTIONS

Chapter Objectives

- Define International Monetary system
- Evolution of International monetary system
- International financial institutions

5. What is the International Monetary System (IMS)?

The international monetary system refers to the operating system of the financial environment, which consists of financial institutions, multinational corporations, and investors. The IMS provides the institutional framework for determining the rules & procedures for international payments, determination of exchange rates, and movement of capital.

An IMS is a set of internationally agreed rules, conventions & supporting institutions that facilitate international trade, cross border investment and generally the reallocation of capital between states that have different currencies.

5.1. Criteria for Evaluating an International Monetary System

IMS are usually classified by two criteria (or along two dimensions): the role of the exchange rate, and the nature of the reserve asset(s).

The role of exchange rate: The need for converting one country's currency into those of others gives the exchange rate regime a prominent place in the IMS. The degree of flexibility of the exchange rate is one basis of classifying IMS. We are already familiar with the permanently fixed & absolutely flexible exchange rates.

Sometimes nations stand ready to change the fixed par value of their currencies, needed to correct fundamental disequilibrium in the BOPs, this system is known as the adjustable peg. A managed float exchange rate regime is one in w/c there are no fixed par values, but the monetary authority does intervene in the foreign exchange market, so as to prevent the exchange rate from

moving far away from its long term trend. The target zone approach-it involves the setting up of a set of mutually consistence targets for (trade-weighted) real exchange rates by major economies (such as U.S, Germany, & Japan). With targets given, the participating countries remain committed to keep the exchange rate within a band (or zone) of 10 per cent on either side of the targets.

The nature of the reserve asset(s): A second way of characterizing the IMS is in terms of the nature of the reserve asset(s). There are two major types of international reserve assets. These are: Commodity reserves, and fiduciary (or fiat) reserves

The essential difference between the two lies in the intrinsic value of the relevant reserve asset. While the commodity reserves (like gold) have some intrinsic value (other than their value as money) fiduciary reserves (such as special Drawing Rights, and national currencies which are not convertible into commodity reserves) have none. On the basis of the nature of reserves held by the monetary authority, a threefold classification of the IMS is possible.

1. **Pure Commodity Standards** : where all reserves consist of commodity reserves (as in the case of gold standard).
2. **Pure Fiduciary Standards:** in which the entire reserve consists of fiduciary reserves (e.g. the inconvertible paper standard).
3. **Mixed Standards:** in which the entire reserves are partly commodity reserves and partly fiduciary reserves(e.g. the gold exchange standard)

5.2. The Evolution of an IMS

The IMS has gone through four stages in its evolution: The gold standard (1870-1914), the gold-exchange standard (1925-1933), the Bretton Woods system (1944-1971), the floating/hybrid exchange rate system (1976-present).

Gold Standard:-under this standard each country fixes the price of its currency in terms of gold by standing ready to trade domestic currency for gold whenever necessary to defend the official price. Because countries tie their currencies to gold under a gold standard, official international reserves take the form of gold. Gold standard rules also require each country to allow unhindered imports & exports of gold across its borders. Under these arrangements, a gold standard, like a

reserve currency system, results in fixed exchange rates b/n all currencies. Example. if the dollar price of gold is pegged at \$35 per ounce by the Federal Reserve while the pound price of gold is pegged at 14.58 pound per ounce by Britain's central bank, the bank of England, the dollar/pound exchange rate must be constant at $(\$35 \text{ per ounce}) \div (£14.58 \text{ per ounce}) = \2.40 per pound .

£ **Benefits and Drawbacks of the Gold Standard**

Benefits: A gold standard places automatic limits on the extent to which central banks can cause increases in national price levels through expansionary monetary policies, this limits make the real values of national monies more stable and predictable, it enhancing the transaction economies arising from the use of money.

Drawbacks: The gold standard places undesirable constraints on the use of monetary policy to fight unemployment, tying currency values to gold ensures a stable overall price level only if the relative price of gold and other goods and Services is stable, an IMS based on gold is problematic because central banks can't increase their holdings of international reserves as their economic grow unless there are continual new gold discoveries.

£ **The Gold Exchange Standard**

Halfway between the gold standard and a pure reserve currency standard is the gold exchange standard. Under a gold exchange standard central banks' reserves consist of gold and currencies whose prices in terms of gold are fixed, and each central bank fixes its exchange rate to a currency with a fixed gold price. A gold exchange standard can operate like a gold standard in restraining excessive monetary growth throughout the world, but it allows more flexibility in the growth of international reserves, which can consist of assets besides gold. A gold exchange standard is, however, subject to the other limitations of a gold standard listed above. The post-world war II reserve currency system centered on the dollar was, in fact, originally setup as a gold exchange standard.

£ **The Bretton Woods System as a Gold Exchange Standard**

The Bretton Woods Agreement was reached in a 1944 summit held in New Hampshire, USA on a site by the same name. A Bretton wood is a system under which the currencies are pegged with

dollar whereas under the gold standard the currencies are pegged to gold. The agreement was reached by 730 delegates, who were the representatives of the 44 allied nations that attended the summit. The delegates, within the agreement, used the gold standard to create a fixed currency exchange rate. The Bretton Woods System is a set of unified rules & policies that provided the framework necessary to create fixed international currency exchange rates.

Essentially, the agreement called for the newly created IMF to determine the fixed rate of exchange for currencies around the world. Every represented country assumed the responsibility of upholding the exchange rate, with incredibly narrow margins above and below. Countries struggling to stay within the window of the fixed exchange rate could petition the IMF for a rate adjustment, w/c all allied countries would then be responsible for following. The Bretton woods system collapsed in August 1971 when U.S President Nixon suspended the gold convertibility of the dollar. This system of fixed exchange rates had two main features: (1) Parity, as the currency of each member country was determined in terms of gold or the dollar, and (2) the price of gold was also fixed in terms of the dollar, which was convertible into gold.

£ The Current System: Managed Float / Hybrid System

Managed Floating exchange rate system: It is an exchange rate system that allows a nation's central bank to intervene regularly in foreign exchange markets to change the direction of the currency's float and/ or reduce the amount of currency volatility. This exchange rate is also known as a "dirty float". Central bank might attempt to bring currency depreciation to: improve the balance of trade or improve the CA by making exports more prices competitive, reduce the risk of a deflationary recession- a lower currency increases export demand & increases the domestic price level by making imports more expensive, rebalance the economy away from consumption towards higher exports & capital investment or to bring about an appreciation of the currency, to curb demand-pull inflationary pressures and to reduce the prices of imported capital & technology or essential inputs to enhance long run growth potential.

Limits to central bank intervention to manage a currency's value: Requires large scale foreign exchange reserves- many smaller and relatively poorer countries do not have these, Central banks intervening on their own may have little or no market power against the sheer(thin) weight of speculative buying and selling in global currency markets (turnover > \$6

trillion a day), and Changing interest rates to influence a currency might conflict against other macroeconomic objectives - raising interest rates to support a currency might stifle(hold back) growth.

5.3. Key International Financial Institutions

An international financial institution (IFI) is a financial institution that has been established by more than one country, and hence is subject to international law. IFIs were established after WWII to assist in the reconstruction of Europe and provide mechanisms for international cooperation in managing the global financial system. Its owners or shareholders are generally national governments, although other international institutions and other organizations occasionally figure as shareholders. The most prominent IFIs are creations of multiple nations, bilateral financial institutions exist and are technically IFIs.

What does the IMF do?

The IMF monitors the international monetary system & global economic developments to identify risks & recommend policies for growth & financial stability. The Fund also undertakes a regular health check of the economic & financial policies of its 190 member countries.

What does the WB do?

The World Bank is an international financial institution that provides loans & grants to the gov'ts of low- & middle-income countries for the purpose of pursuing capital projects. Their approaches to achieving this shared goal are complementary: the IMF focuses on macroeconomic & financial stability while the WB concentrates on long-term economic development and poverty reduction.

5.4. International Capital Flows and Multinational Corporations

Multinational corporations leverage their financial position and access to global markets to raise capital in a cost-effective and efficient manner. This gives these companies an advantage over small domestic operators that do not have the same level of credit or cash, but there are risks associated with international finance. The capital structure multinationals use directly impacts profitability, growth and sustainability.

Invested Capital: A multinational's capital structure comprises the sources of money used to finance operations, expand production or purchase assets. Companies acquire capital through the sale of securities in financial markets such as the New York Stock Exchange or the London Stock Exchange. Debt and equity are the two forms of capital that multinationals have to choose from, and each form has its advantages and disadvantages. The cost of raising capital is an important component of financing decision.

Debt Financing: Acquiring debt capital is a process that is contingent on the availability of funds in the global credit markets, interest rates and a corporation's existing debt obligations. If credit markets are experiencing a contraction, it may be difficult for the corporation to sell corporate bonds at favorable rates. In particular, it may be challenging to get high advance rates for asset-backed securities. If a firm becomes over-leveraged, it may be unable to pay its debt obligations leading to insolvency. However, debt costs less to acquire than other forms of financing.

Equity Financing: Preferred stock, common stock and components of retained earnings are considered equity capital. It is important for a multinational to carefully analyze its equity cash flows and mitigate the risk associated with currency fluctuations. Otherwise, it may lose equity due to changes in exchange rates. Also, the issuance of new shares may cause stock prices to fall because investors no longer feel company shares are worth their pre-issuance price. Offering stock in global financial markets costs multinationals more than acquiring debt, but it may be the right financing option if a corporation is already highly leveraged.

Tax Considerations: Multinationals have the option to shift income to jurisdictions where the tax treatment is the most advantageous. As a result, debt and equity financing decisions are different relevant to solely domestic companies. If income is reported in the United States, it may be beneficial to obtain debt financing, because the interest is tax-deductible. When making capital structure decisions, multinationals must evaluate their tax planning strategies to minimize their tax liabilities.

5.5. The concept of Dutch Disease

What is Dutch Disease?

Dutch disease is a concept that describes an economic phenomenon where the rapid development of one sector of the economy (particularly natural resources) precipitates a decline in other sectors. It is also often characterized by a substantial appreciation of the domestic currency. Dutch disease is a paradoxical situation where good news for one sector of the economy, such as the discovery of natural resources, results in a negative impact on the country's overall economy.

£ Breaking Down the Dutch Disease Phenomenon

The Dutch disease term was first introduced in The Economist magazine in 1977 to analyze the economic situation in the Netherlands (hence the name) after the discovery of large natural gas fields in 1959. Although the Dutch economy increased its revenues from the export of natural gas, the significant appreciation of the national currency from the large capital influx into the sector resulted in a higher unemployment rate in the country, as well as a decline in the manufacturing industry.

The phenomenon of Dutch disease commonly occurs in countries whose economies rely heavily on the export of natural resources. The paradox contradicts the concept of comparative advantage. According to the comparative advantage model, each country should specialize in the industry in which it possesses a comparative advantage over other countries.

However, it does not work well with countries that primarily export natural resources. For example, the volatility of commodity prices cannot sustain a country's economy for long time periods. Also, the overdependence on the export of natural resources leads to the underdevelopment of other sectors of the economy, such as manufacturing and agriculture.

£ The Workings of Dutch Disease

The negative influence of Dutch disease on the economy can be explained by some features attributable to the sectors that are related to natural resources. For example, mining industries generally require heavy capital investments, but they are not labor-intensive. Therefore, multinational corporations and foreign countries that have capital are often interested in investing in such ventures.

Foreign investment may lead to higher demand for the country's domestic currency, and it will start appreciating. The appreciation of the domestic currency will make the country's exports in other industries more expensive while imports will become cheaper.

Subsequently, domestic producers will face lower demand for their products abroad, as well as greater competition from foreign producers. Thus, the lagging sectors of the economy will face further troubles.

How to Avoid Dutch Disease?

The two primary strategies that can help solve Dutch disease are listed below:

- 1. Deceleration of domestic currency appreciation:** The deceleration of currency appreciation is an easier and more viable strategy to prevent the adverse effects of Dutch disease. It can sometimes be achieved by smoothing the spending of revenues earned from the export of natural resources. One of the most common methods to do so is the creation of a sovereign wealth fund. Many developed and developing countries, including Australia, Canada, Norway, and Russia, manage large sovereign wealth funds. Sovereign wealth funds aim to stabilize the inflows of capital into the economy to prevent it from overheating and causing significant currency appreciation. Excess revenues can be spent on education or infrastructure that will help to diversify the economy.
- 2. Diversification of the economy:** The diversification of the economy is a strategy that can almost eliminate the negative impact of Dutch disease on the economy. Economic diversification can be achieved by subsidizing lagging sectors of the economy or establishing tariffs to support domestic producers.

Review Questions

1. Explain Bretton Woods System of exchange rate? How it was set up and maintained?

2. What was decided by the group of ten under the Smithsonian Agreement in December 1971?
3. What was the main cause of Dutch disease? How do you explain it?
4. What is special drawing right (SDR)? How do nations apply SDRs?